Scattering Amplitudes in 6D SQCD and 6D Supergravity

Venkat Bharadwaj Sekaran

Supervisor: Henrik Johansson

Subject Reader: Chia-Hsien Shen
Examiner: Dmytro Volin

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UPPSALA
UNIVERSITET

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Abstract

Finding modern techniques that bypass the need for Feynman rules has played a crucial role in simplifying the complex calculations involved in computing scattering amplitudes at higher orders and/or high multiplicities. 6D supersymmetric theories have a rich structure while admitting dimensional reduction to known 4D theories, making them a natural setting to study scattering amplitudes. In this thesis, we examine techniques for constructing tree-level scattering amplitudes in supersymmetric 6D theories. We start by reviewing spinor helicity, on-shell recursion relations, and supersymmetry in 6D. Using known expressions from 6D and 5D supersymmetric theories, we construct the 3,4-point superamplitudes in the $\mathcal{N}$=(1,0) super quantum chromodynamics theory. As an alternative to on-shell recursion relations, we introduce a bootstrap approach that uses supersymmetric constraints and amplitude relations, which is then used to construct a 5-point amplitude in the SQCD theory. Using the double copy structure, the corresponding 3,4,5-point superamplitudes are constructed in the $\mathcal{N}$=(2,0) chiral and $\mathcal{N}$=(1,1) non-chiral supergravity theories. Finally, we also provide explicit maps from the 6D theories to 4D $\mathcal{N} = 2$ SQCD and $\mathcal{N} = 4$ supergravity theories that can be used to construct massive superamplitudes in 4D from the 6D amplitudes.

Populärvetenskaplig sammanfattning

Ett av fysikens huvudmål är att förstå dynamiken vid kollisioner mellan elementarpartiklar. Förutsägelser om dessa görs med hjälp av spridningsamplituder, vilka beskriver sannolikheten för specifika utfall av sådana kollisioner. Dessa storheter beräknas vanligtvis med hjälp av Feynmandiagram. Antalet diagram ökar dramatiskt beroende på hur många partiklar som är inblandade och vilken beräkningsprecision som krävs. Särskilt i teorier som försöker beskriva gravitation kan dessa diagram vara mycket komplexa. Fysiker arbetar därför aktivt med att hitta alternativa metoder som förenklar beräkningarna genom att undvika behovet av Feynmandiagram. Ett sätt är att införa symmetrier i en teori; detta ökar antalet villkor man kan utgå ifrån, vilket gör beräkningarna enklare. Supersymmetri är en sådan symmetri som introducerar partnerpartiklar till kända elementarpartiklar. ”Double copy”-metoden är en annan teknik som används för att beräkna amplituder inom gravitationsteorier utifrån amplituder i gaugeteorier (såsom teorin som beskriver dynamiken hos kvarkar eller fotoner), vilka är relativt enklare att beräkna.

I den fyrdimensionella världen (tre rumsdimensioner och en tidsdimension) vi uppfattar har de flesta partiklar massa, vilket försvårar beräkningen av amplituder. Genom att studera teorier i högre dimensioner kan vi dock betrakta samma partiklar som masslösa, vilket avsevärt förenklar beräkningarna. Rörelsen i de extra dimensionerna kan, när den reduceras till fyra dimensioner (genom en metod som kallas kompaktifiering), kopplas till partiklarnas massa i fyra dimensioner. I denna avhandling studerar vi metoder för att beräkna spridningsamplituder inom 6D-supersymmetrisk kvantkromodynamik (en teori som beskriver kvarkar, gluoner och deras superpartner) samt inom 6D-supergravitationsteorier. Vi börjar med att konstruera spridningsamplituder för tre, fyra och fem partiklar i 6D-SQCD med hjälp av moderna metoder som utnyttjar teorins supersymmetri och partiklarnas masslösa natur. Därefter konstruerar vi motsvarande amplituder inom 6D-supergravitation med hjälp av ”double copy”-strukturen. Slutligen presenterar vi en koppling från dessa masslösa 6D-teorier till kända 4D-teorier; denna kan användas för att erhålla realistiska spridningsamplituder som i sin tur kan användas för att studera dynamiken hos massiva partiklar som påverkas av gravitation. Av särskilt intresse är fall där dessa massiva partiklar kan modelleras som svarta hål eller neutronstjärnor, vilket kommer att ge värdefull information för mätningar av gravitationsvågor.

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1 Introduction

Calculating scattering amplitudes has been a topic of great importance for the development of physics. These quantities have been a bridge from theoretical results to observable quantities, while also being uniquely determined for a specified quantum field theory [1, 2]. Traditional methods of calculating scattering amplitudes using Feynman diagrams involve tedious computations, especially for complicated theories that involve gravitational interactions. In recent decades, researchers have been developing modern amplitude techniques that simplify these computations [1, 3, 4].

Spinor helicity formalism is one such technique developed to compute 4D massless amplitudes [1, 5]. Since then, this formalism has been extended to other dimensions [6–9], as well as to massive amplitudes [7, 10]. In particular, a detailed review of spinor helicity for massless states in 6D is presented in [8]. On-shell recursion relations such as the Britto-Cachazo-Feng-Witten (BCFW) recursion relations are another avenue that helps construct amplitudes of higher multiplicities from smaller amplitudes [8, 11].

Supersymmetric theories are a subset of quantum field theories that are of special interest due to their rich structure. These theories provide additional constraints on scattering amplitudes, which simplifies their computation [1, 12]. In 4D, the $\mathcal{N} = 4$ super Yang-Mills (SYM) has been studied extensively due to its maximal supersymmetry and conformal properties [13].

Studying such theories in higher dimensions provides richer structure, while also encoding information about lower-dimensional amplitudes. Compactification maps massless particles in higher dimensions to massive particles in lower dimensions, so massless spinor helicity in higher dimensions offers a cleaner alternative to massive spinor helicity in lower dimensions. In particular, the 6D $\mathcal{N} = (1, 1)$ SYM has also been studied [14]. The amplitudes in this theory have also been used to compute amplitudes in 4D $\mathcal{N} = 4$ SYM by dimensional reduction [15].

Supergravity theories in 4D have also been important to study as a consistent quantum theory of gravity. Although computing amplitudes in these theories can become more complicated than in gauge theories [16, 17], with the presence of the double-copy structure, the amplitudes can be computed from gauge theory results [17, 18]. Building on these, research on supergravity theories has also extended to higher dimensions. In particular, 6D $\mathcal{N} = (2, 0)/\mathcal{N} = (1, 1)$ supergravity theories are of great interest due to their surprising UV-finite nature at one loop [19].

While the maximally supersymmetric theories have been studied extensively, the half maximal theories prove to be interesting grounds for research due to their less constrained nature [20]. $\mathcal{N} = 2$ super quantum chromodynamics (SQCD) is one such theory, with its low-energy dynamics exactly determined [21]. In this thesis, we study techniques to compute tree-level amplitudes in the half-maximal 6D $\mathcal{N} = (1, 0)$ super quantum chromodynamics (SQCD) theory and 6D $\mathcal{N} = (2, 0)/\mathcal{N} = (1, 1)$ supergravity using on-shell relations, the bootstrap approach, and double copy structure. We also discuss the reduction of these 6D theories to the 4D massive $\mathcal{N} = 2$ SQCD and $\mathcal{N} = 4$ supergravity theories.

The thesis is structured as follows: section 2 introduces the SQCD theory in 6D. Sections 3 and 4 discuss spinor helicity in 4D and 6D, and on-shell recursion relations, respectively. In section 5, supersymmetry is introduced, and 3,4-point superamplitudes in 6D $\mathcal{N} = (1, 0)$ SQCD are computed. Section 6 describes a bootstrap approach to compute the 5-point superamplitude. In section 7, the 3,4,5-point superamplitudes in 6D $\mathcal{N} = (2, 0)$ and $\mathcal{N} = (1, 1)$ supergravity theories are calculated. Section 8 provides a map from these 6D amplitudes onto 4D theories. The thesis ends with a conclusion that also talks about possible future work.

2 Lagrangian

Before we start computing amplitudes, we need to establish the Lagrangian that governs the 6D super quantum chromodynamics (SQCD) theory. Similar to the 4D QCD theory, we first start with the gauge fields (A^a_μ) and the fermion (quark) fields ($\bar{\psi}, \psi$)

$$\mathcal{L} = -\frac{1}{4} F^a_{\mu\nu} F^{\mu\nu}_a + \bar{\psi}(i\Gamma^\mu D_\mu)\psi, \tag{2.1}$$

where a is the gauge group $SU(N_c)$ index, and

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu, \tag{2.2a}$$
$$D_\mu = (\partial_\mu - igA^a_\mu T^a). \tag{2.2b}$$

This Lagrangian contains 4 bosonic degrees of freedom that transform under the adjoint action of the gauge group, and 4 ($2 \times 2^3 = 16$ real DoF in 6D, applying Weyl projection and on-shell Dirac condition $\implies 16/(2 \times 2) = 4$) fermionic degrees of freedom that transform under fundamental (anti-fundamental) action of the gauge group. To introduce supersymmetry for the SQCD theory, we introduce gaugino (fermionic partner of gauge boson) fields ($\bar{\lambda}^a_1, \lambda^a_1, \bar{\lambda}^a_1, \lambda^a_1$) and scalar(bosonic partner for matter fermions) fields ($\phi_1, \phi_2, \bar{\phi}_2, \bar{\phi}_2$). With the addition of these fields, the kinetic terms in the Lagrangian can be written as

$$\mathcal{L}_{\text{kin}} = -\frac{1}{4} F^a_{\mu\nu} F^{\mu\nu}_a + \frac{i}{2} \bar{\lambda}^a_i (i\Gamma^\mu D_\mu)\lambda^{ia} - D_\mu \phi_i D^\mu \bar{\phi}^i + \bar{\psi}(i\Gamma^\mu D_\mu)\psi. \tag{2.3}$$

Counting the degrees of freedom (DoF), for particles under the adjoint action of the gauge group, the vector fields, it seems like there are 8 DoFs from the gauginos. However, due to the pseudo-reality condition in 6D ($\lambda_i = \epsilon_{ij} \mathcal{C} \bar{\lambda}^j$), the DoF gets cut in half back to 4.

$$A^a_\mu \implies 6 - 2 = 4 \text{ bosonic DoF}; \quad \{\bar{\lambda}^a_i, \lambda^a_i\} \implies 4 \text{ fermionic DoF}. \tag{2.4}$$

And for particles that transform under the fundamental representation of the gauge group, the matter fields,

$$\{\phi_i, \bar{\phi}_i\} \implies 4 \text{ bosonic DoF}; \quad \{\bar{\psi}, \psi\} \implies 4 \text{ fermionic DoF}. \tag{2.5}$$

Finally, the interaction Lagrangian that is allowed by the supersymmetry is given as

$$\mathcal{L}_{\text{int}} = (ig\bar{\psi}\lambda_i \phi^i - ig\bar{\lambda}_i \psi \phi^i) - \frac{g^2}{2} (\bar{\phi}_i T^a \phi^i)^2. \tag{2.6}$$

Thus, the total 6D SQCD Lagrangian is given by

$$\mathcal{L}^{\text{6D}}_{\text{SQCD}} = \mathcal{L}_{\text{kin}} + \mathcal{L}_{\text{int}}. \tag{2.7}$$

2.1 Color

Having multiple copies of the fields allows the theory to have a symmetry that rotates them into each other. The charge corresponding to this symmetry is called the color. In the 4D standard model, the QCD theory has three color charges and has the gauge group $SU(3)$. For this theory, the gauge (color) group is $SU(N_c)$. The generators (T^a) of this group obey the relations.

$$\text{Tr}(T^a T^b) = \delta^{ab}; \quad [T^a, T^b] = f^{abc} T^c. \tag{2.8}$$

A typical n-point in a gauge theory can be written as

$$\mathbf{A}_n^{\text{tree}} = g^{n-2} \sum_{\sigma \in S_{n-2}} A_n^{\text{tree}}(1, \sigma, n) \text{Tr}(T^1 T^{\sigma(1)} \cdots T^n). \tag{2.9}$$

Throughout this thesis, unless mentioned otherwise, we will only deal with the color-ordered partial amplitudes A_n^{tree}.

2.2 Flavor

Similar to the QCD theories in 4D, it is also possible for the theory to have multiple sets of the matter fields of the same color. For N_f copies of such particles, this theory also exhibits flavor symmetry under the $USp(2N_f)$. For such a theory, the defining symplectic form is $\Omega_{f_i f_j}$ which satisfies the properties

$$\Omega_{f_i f_j} = -\Omega_{f_j f_i}, \tag{2.10a}$$

$$U^{f_i f_j} \Omega_{f_j f_k} (U^T)^{f_k f_l} = \Omega^{f_i f_l}. \tag{2.10b}$$

3 Spinor helicity formalism

Spinor helicity formalism is a tool introduced in 4D to construct on-shell scattering amplitudes from little group[1] covariant objects [1]. This is a result of the fact that the states of massless particles are uniquely determined by their momentum and helicity. The advantage of this approach is that we can compute amplitudes without any mention of gauge fixing or polarization vectors.

In this section we review the spinor helicity formalism in 4D and 6D, and define some building blocks alongside their useful properties.

3.1 4D

For massless particles in 4D, consider a reference momentum such as $p_\mu = (p_0, p_0, 0, 0)$. Any rotation in the bottom two spatial axes leaves the momentum invariant. Thus, the little group of the massless particles is $SO(2) \cong U(1)$. Similarly, for a massive particle in 4D, considering $p_\mu = (m, 0, 0, 0)$ as the reference momentum, the little group can be identified as $SO(3)$.

For a massless particle, consider the object

$$p_\mu \sigma^\mu_{\alpha\dot{\alpha}} = p_{\alpha\dot{\alpha}} \tag{3.1}$$

where $\sigma^\mu_{\alpha\dot{\alpha}}$ are the 4D Pauli matrices. Since $p_\mu p^\mu = 0 \implies \det(p_{\alpha\dot{\alpha}}) = 0$, $p_{\alpha\dot{\alpha}}$ is a null matrix, the four-momentum can be represented as a bispinor of the form

$$p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}} \tag{3.2}$$

where λ_α and $\tilde{\lambda}_{\dot{a}}$ are complex-valued spinors and the $\alpha, \dot{\alpha}$ are the $SL(2, \mathbb{C})$ indices that come from the spin representation of the Lorentz group SO(1,3). Solving the massless Weyl equation for momentum p_μ gives us precisely the spinors that can be used to represent the momentum.

[1] The subgroup of the Lorentz group consisting of transformations that leave a specific momentum invariant.

The action of the little group, $U(1)$, on the spinors are

$$\lambda_\alpha \to t\lambda_\alpha; \quad \tilde{\lambda}_{\dot{\alpha}} \to t^{-1}\tilde{\lambda}_{\dot{\alpha}}; \quad t \in \mathbb{C}. \tag{3.3}$$

The action of the little group on these spinors is fixed, so we omit a little group index on these spinors. One convenient form of writing these spinors is using angle and square brackets as

$$\lambda_{i\alpha} = |i\rangle_\alpha\,, \tag{3.4a}$$
$$\tilde{\lambda}_{i\dot{\alpha}} = [i|_{\dot{\alpha}}, \tag{3.4b}$$

where i is an index introduced to distinguish particles.

The next step is to construct objects that are Lorentz-invariant building blocks starting from these spinors. We can contract two α indices with a 2D Levi-Cevita tensors ($\epsilon_{\alpha\beta}, \epsilon_{\dot{\alpha}\dot{\beta}}$) which are $SL(2,\mathbb{C})$ covariant objects. Thus, we have two invariants

$$\langle ij\rangle := \lambda_{i\alpha}\lambda_{j\beta}\epsilon^{\alpha\beta}, \tag{3.5a}$$
$$[ij] := \lambda_{i\dot{\alpha}}\lambda_{j\dot{\beta}}\epsilon^{\dot{\alpha}\dot{\beta}}. \tag{3.5b}$$

Using these objects, we can now construct the 3-point amplitude using just little group scaling arguments.

3-point amplitude

For a general 3-point amplitude, consider three massless particles with helicities h_1, h_2, h_3 and momentum p_1, p_2, p_3. From momentum conservation,

$$\begin{aligned} & p_1 + p_2 + p_3 = 0 \\ \implies & (p_1 + p_2 + p_3)^2 = 0 \\ \implies & 2(p_1 \cdot p_2 + p_2 \cdot p_3 + p_3 \cdot p_1) = 0 \\ \implies & \langle 12\rangle[12] + \langle 23\rangle[23] + \langle 31\rangle[31] = 0. \end{aligned} \tag{3.6}$$

Thus, either $\langle ij\rangle = 0$ or $[ij] = 0$. Using this, we can write a general ansatz for the 3-point amplitude as

$$A(1_{h_1}2_{h_2}3_{h_3}) = c\langle 12\rangle^a\langle 23\rangle^b\langle 31\rangle^b + c'[12]^{a'}[23]^{b'}[31]^{c'} \tag{3.7}$$

Since we want the amplitude to be covariant under little group transformation, we have the exponents fixed as

$$\begin{aligned} & h_1 = -\frac{\alpha+\gamma}{2}; \quad h_2 = -\frac{a+b}{2}; \quad h_3 = -\frac{b+c}{2} \\ \implies & a = -h_1 + h_3 - h_2; \quad b = -h_2 + h_1 - h_3; \quad c = -h_3 + h_2 - h_1. \end{aligned} \tag{3.8}$$

Similarly,

$$\begin{aligned} & h_1 = \frac{a'+c'}{2}; \quad h_2 = \frac{a'+b'}{2}; \quad h_3 = \frac{b'+c'}{2} \\ \implies & a' = h_1 - h_3 + h_2; \quad b' = h_2 - h_1 + h_3; \quad c' = h_3 - h_2 + h_1. \end{aligned} \tag{3.9}$$

Thus, we have a general expression for any 3-point amplitude as follows:

$$\begin{aligned} A(1_{h_1}2_{h_2}3_{h_3}) =& c\langle 12\rangle^{-(h_1-h_3+h_2)}\langle 23\rangle^{-(h_2-h_1+h_3)}\langle 31\rangle^{-(h_3-h_2+h_1)} \\ &+ c'[12]^{(h_1-h_3+h_2)}[23]^{(h_2-h_1+h_3)}[31]^{(h_3-h_2+h_1)}. \end{aligned} \tag{3.10}$$

However, we note that the dimension of the constant c is determined from the theory. Thus, by dimensional analysis, it will be straightforward to see whether the amplitudes depend on $\langle ij\rangle$ or $[ij]$ or if they vanish.

For example, in the YM theory, the coupling constant is dimensionless, and the 3-point interactions are of the form $AA\partial A$, i.e., have a mass dimension of 1. Thus, the amplitude has mass dimension 1 (from the vertex). Then, the three gluon amplitudes are given as

$$\begin{aligned} A(g_1^- g_2^- g_3^+) &= g\langle 12\rangle^3\langle 23\rangle^{-1}\langle 31\rangle^{-1}, \\ A(g_1^+ g_2^+ g_3^-) &= g[12]^3[23]^{-1}[31]^{-1}, \\ A(g_1^+ g_2^+ g_3^+) &= A(g_1^- g_2^- g_3^-) = 0. \end{aligned} \tag{3.11}$$

3.2 6D

In 2009, the spinor helicity formalism was extended to 6D by C. Cheung and D. O'Connell [8]. Similar to the 4D formalism, the goal is to represent the momenta of massless particles as bispinors. These bispinors have an $SU^*(4)$ (spin representation of the Lorentz group $SO(1,5)$) index and a little group index. In 6D, the little group for the massless particles is $SO(4) \cong SU(2)_L \times SU(2)_R$. Then, the momentum is represented as

$$p^{AB} = p_\mu \bar{\Sigma}^{\mu AB} = \lambda_a^A \lambda_b^B \epsilon^{ab}, \tag{3.12}$$

$$p_{AB} = p_\mu \Sigma^\mu_{AB} = \tilde{\lambda}_{\dot{a}A}\tilde{\lambda}_{\dot{b}B}\epsilon^{\dot{a}\dot{b}}, \tag{3.13}$$

where $\Sigma^\mu, \bar{\Sigma}^\mu$ are generalizations of the Pauli matrices in 6D, A, B are $SU^*(4)$ indices, a, b are $SU(2)_L$ indices and $\dot{a}, \dot{b}$ are $SU(2)_R$ indices.

The $SU(2)$ indices can be freely raised or lowered using the 2D Levi-Cevita tensor. While the Lorentz spin index on the spinors cannot be raised or lowered freely, the 4D Levi-Cevita tensor ϵ_{ABCD} (invariant $SU^*(4)$ object) can be used to raise or lower the indices on the momentum giving a relation between them as

$$p_{AB} = \frac{1}{2}\epsilon_{ABCD}p^{CD}. \tag{3.14}$$

To simplify the notation, we introduce the angle and square brackets to represent the chiral and anti-chiral spinors as

$$|i_a\rangle^A := \lambda_{ia}^A, \tag{3.15a}$$

$$|i_a]_A := \tilde{\lambda}_{i\dot{a}A}, \tag{3.15b}$$

where i is the particle index.

By solving the 6D Weyl equation, we can get an explicit parametrization of these spinors. Up to a normalization factor between the columns, the spinors for momentum $p^\mu = (p_0, p_1, p_2, p_3, p_4, p_5)$

are given as [15]

$$|p_a\rangle^A = \begin{pmatrix} \sqrt{p_0+p_3} & 0 \\ \frac{p_1+ip_2}{\sqrt{p_0+p_3}} & \frac{p_5-ip_4}{\sqrt{p_0+p_3}} \\ \frac{-p_5-ip_4}{\sqrt{p_0+p_3}} & \frac{p_1-ip_2}{\sqrt{p_0+p_3}} \\ 0 & -\sqrt{p_0+p_3} \end{pmatrix}, \tag{3.16a}$$

$$|p^{\dot{a}}]_A = \begin{pmatrix} -\sqrt{p_0-p_3} & 0 \\ \frac{p_1-ip_2}{\sqrt{p_0-p_3}} & \frac{p_5+ip_4}{\sqrt{p_0-p_3}} \\ \frac{-p_5+ip_4}{\sqrt{p_0-p_3}} & \frac{p_1+ip_2}{\sqrt{p_0-p_3}} \\ 0 & \sqrt{p_0-p_3} \end{pmatrix}. \tag{3.16b}$$

As a consequence of Eq.(3.14), these spinors are related as,

$$\lambda_a^A \lambda^{aB} = \frac{1}{2}\epsilon^{ABCD}\tilde{\lambda}_{\dot{a}C}\tilde{\lambda}^{\dot{a}}_D.$$

Angle, square brackets and their properties

With these spinors, we can now construct Lorentz-invariant objects that will act as building blocks for the scattering amplitudes. One possible object is obtained by contracting the $SU(4)$ index of a chiral spinor with an anti-chiral spinor as

$$\langle i_a|j_{\dot{b}}] := \lambda_a^A \tilde{\lambda}_{\dot{b}A}. \tag{3.17}$$

The Mandelstam variables are also Lorentz invariant objects and can be represented using the spinors as

$$s_{ij} := (p_i+p_j)^2 = 2p_i\cdot p_j = \frac{1}{2}\epsilon_{ABCD}p_i^{AB}p_j^{CD} = -\det\left(\langle i_a|j_{\dot{b}}]\right). \tag{3.18}$$

Another invariant object is obtained by contracting four chiral (anti-chiral) spinors with the 4D Levi-Civita tensor

$$\langle i_a j_b k_c l_d\rangle := \epsilon_{ABCD}\,|i_a\rangle^A\,|j_b\rangle^B\,|k_c\rangle^C\,|l_d\rangle^D\,, \tag{3.19a}$$

$$[i_{\dot{a}} j_{\dot{b}} k_{\dot{c}} l_{\dot{d}}] := \epsilon^{ABCD}|i_{\dot{a}}]_A|j_{\dot{b}}]_B|k_{\dot{c}}]_C|l_{\dot{d}}]_D. \tag{3.19b}$$

Since the spin group $SU(4)$ allows for at most four linearly independent spinors, another property satisfied by these spinors is the Schouten identity in 6D. Using the objects defined above, the identity is written as

$$\langle 1234\rangle\,|5\rangle + \langle 2345\rangle\,|1\rangle + \langle 3451\rangle\,|2\rangle + \langle 4512\rangle\,|3\rangle + \langle 5123\rangle\,|4\rangle = 0, \tag{3.20a}$$

$$[1234]|5] + [2345]|1] + [3451]|2] + [4512]|3] + [5123]|4] = 0. \tag{3.20b}$$

Other invariants that can be constructed by contracting the $SU(4)$ indices are

$$\langle i_a|\not{p}_{a_1}\cdots\not{p}_{a_{2n-1}}|j_b\rangle := \lambda_{ia}^{A_1} p_{a_1A_1A_2} p_{a_2}^{A_2A_3}\cdots p_{a_{2n-1}A_{2n-1}A_{2n}}\lambda_{jb}^{A_{2n}}, \tag{3.21a}$$

$$\langle i_a|\not{p}_{a_1}\cdots\not{p}_{a_{2n}}|j_{\dot{b}}] := \lambda_{ia}^{A_1} p_{a_1A_1A_2} p_{a_2}^{A_2A_3}\cdots p_{a_{2n}}^{A_{2n}A_{2n+1}}\tilde{\lambda}_{A_{2n+1}j\dot{b}}, \tag{3.21b}$$

where the $\not{p}$ is defined to be the momentum p_μ contracted with an appropriate sigma matrix $(\Sigma^{\mu AB},\, \bar{\Sigma}^{\mu}_{AB})$.

Polarization vector

The polarization vector is an important object used to construct scattering amplitudes using Feynman rules. While we don't need them to construct amplitudes using spinor helicity, it is good to see how the object is written in this formalism. For a particle with momentum p, the polarization vector is defined as

$$\begin{aligned}\varepsilon^{\mu}_{a\dot{a}} =&\langle p_a|\Sigma^{\mu}|q_b\rangle(\langle q_b|p^{\dot{a}}]\rangle)^{-1} = \frac{1}{s_{pq}}\langle p_a|\Sigma^{\mu}|q_b\rangle\langle q^b|p_{\dot{a}}] \\ =&\frac{1}{s_{pq}}[p_{\dot{a}}|\bar{\Sigma}^{\mu}|q_{\dot{b}}][q^{\dot{b}}|p_a\rangle,\end{aligned} \tag{3.22}$$

where q is a reference null-momentum such that $p \cdot q \neq 0$. The polarization vector in 6D transforms under the little group $SU(2) \times SU(2)$. The choice of q here can be translated as a gauge choice for our vector. Thus, changing q will result in a change in $\varepsilon^{\mu}_{a\dot{a}}$ that is proportional to the momentum p. This definition also satisfies the transversality property, $p_{\mu}\varepsilon^{\mu}_{a\dot{a}} = 0$.

3.3 Three gluon amplitude

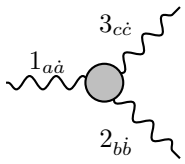


Figure 1: 3-point Feynman diagram for three gluons. Vector particles (gluons here) are represented using wavy lines along with the particle number and little group dependence.

Now that the basic objects and their properties are defined, we can compute a general 3-point amplitude. Since each particle (as shown in Fig. 1) is defined by two little group indices, we can construct an ansatz as,

$$A(1_{a\dot{a}}2_{b\dot{b}}3_{c\dot{c}}) \stackrel{?}{=} \langle 1_a|2_{\dot{b}}]\langle 2_b|3_{\dot{c}}]\langle 3_c|1_{\dot{a}}]$$

Here, we would need a prefactor with mass dimensions -2 for the amplitude to have mass dimension 1. So, one would need the inverses of the building blocks $\langle i_a|j_{\dot{b}}]$. However, as we saw in the 4D case Eq.(3.6), kinematic invariants $p_i \cdot p_j = \det(\langle i_a|j_{\dot{b}}])$ vanish due to momentum conservation and therefore these blocks are singular.

In [8], this is resolved by using that $\langle i_a|j_{\dot{b}}]$ are rank 1 matrices and can be constructed as products of two spinors such that $\langle i_a|j_{\dot{b}}] = u_{ia}\tilde{u}_{j\dot{b}}$. Using this, one can define inverse spinors w's that satisfy

$$\begin{aligned}u_{ia}w_{ib} - u_{ib}w_{ia} &= \epsilon_{ab}, \\ \tilde{u}_{i\dot{a}}\tilde{w}_{i\dot{b}} - \tilde{u}_{i\dot{b}}\tilde{w}_{i\dot{a}} &= \epsilon_{\dot{a}\dot{b}}.\end{aligned} \tag{3.23a}$$

Before constructing the amplitude, we study some constraints on these spinors. Starting from momentum conservation, we have

$$p_1^{AB} + p_2^{AB} + p_3^{AB} = 0$$

$$
\begin{aligned}
&\implies |1_a\rangle\langle 1^a| + |2_b\rangle\langle 2^b| + |3_c\rangle\langle 3^c| = 0;\\
&\implies [1_{\dot a}|1_a\rangle\langle 1^a| + [1_{\dot a}|2_b\rangle\langle 2^b| + [1_{\dot a}|3_c\rangle\langle 3^c| = 0\\
&\implies u_{2b}\langle 2^b| = u_{3c}\langle 3^c|. \qquad (3.24)
\end{aligned}
$$

Repeating this, we find that

$$
\begin{aligned}
u_{2b}\langle 2^b| = u_{3c}\langle 3^c| = u_{1a}\langle 1^a|,\\
\tilde u_{3\dot c}[3^{\dot c}| = \tilde u_{2\dot b}[2^{\dot b}| = \tilde u_{1\dot a}[1^{\dot a}|. \qquad (3.25)
\end{aligned}
$$

Similarly, to find constraints on the w's, we use the definition Eq.(3.23a) to relate them to the momenta as

$$
\begin{aligned}
&u_{ia}w_{ib} - u_{ib}w_{ia} = \epsilon_{ab}\\
\implies &p_i^{AB} = |i^a\rangle\epsilon_{ab}\langle i^b|\\
\implies &p_i^{AB} = |i^a\rangle(u_{ia}w_{ib} - u_{ib}w_{ia})\langle i^b|\\
\implies &p_i^{AB} = |i.u_i\rangle\langle w_i.i| - |i.w_i\rangle\langle u_i.i| \qquad (3.26)
\end{aligned}
$$

with $|i.u_i\rangle = |i^a\rangle u_{ia}$. Then using momentum conservation,

$$
\begin{aligned}
&\implies |1.u_1\rangle\langle w_1.1| - |1.w_1\rangle\langle u_1.1| + |2.u_2\rangle\langle w_2.2| - |2.w_2\rangle\langle u_2.2| + |3.u_3\rangle\langle w_3.3| - |3.w_3\rangle\langle u_3.3| = 0\\
&\implies |1.u_1\rangle(\langle w_1.1| + \langle w_2.2| + \langle w_3.3|) - (|1.w_1\rangle + |2.w_2\rangle + |3.w_3\rangle)\langle u_1.1| = 0\\
&\implies |1.w_1\rangle + |2.w_2\rangle + |3.w_3\rangle = 0. \qquad (3.27)
\end{aligned}
$$

However, this constraint on the inverses also allows the transformation $w_i \to w_i + b_i u_i$ with $\sum_i b_i = 0$ (from Eq.(3.25)). Thus, when we construct the amplitude, we necessitate that the amplitudes must also be invariant under such translations. Now that we have their properties, we can construct the 3-point amplitude with $u_a(\tilde u_{\dot a})$ as objects of mass-dimension $1/2$ and $w_a(\tilde w_{\dot a})$ as objects of mass-dimension $-1/2$, each with a little group index.

For the 3-point amplitude, we want three $SU(2)_L$ and three $SU(2)_R$ little group indices with total mass-dimension 1. Consider a tensor as follows:

$$
\Gamma_{abc} = u_{1a}u_{2b}w_{3c} + u_{1a}w_{2b}u_{3c} + w_{1a}u_{2b}u_{3c}. \qquad (3.28)
$$

Under $b-$translation,

$$
\begin{aligned}
\Gamma_{abc} =&u_{1a}u_{2b}w_{3c} + u_{1a}w_{2b}u_{3c} + w_{1a}u_{2b}u_{3c}\\
&+ b_3u_{1a}u_{2b}u_{3c} + b_2u_{1a}u_{2b}u_{3c} + b_1u_{1a}u_{2b}u_{3c}\\
=&u_{1a}u_{2b}w_{3c} + u_{1a}w_{2b}u_{3c} + w_{1a}u_{2b}u_{3c}\\
&+ (b_1 + b_2 + b_3)u_{1a}u_{2b}u_{3c}\\
=&u_{1a}u_{2b}w_{3c} + u_{1a}w_{2b}u_{3c} + w_{1a}u_{2b}u_{3c},
\end{aligned}
$$

i.e., it is invariant. Similarly, one can construct the tensor

$$
\Gamma_{\dot a\dot b\dot c} = \tilde u_{1\dot a}\tilde u_{2\dot b}\tilde w_{3\dot c} + \tilde u_{1\dot a}\tilde w_{2\dot b}\tilde u_{3\dot c} + \tilde w_{1\dot a}\tilde u_{2\dot b}\tilde u_{3\dot c}. \qquad (3.29)
$$

Together, they have mass-dimensions 1 (2 from $u\tilde u$'s and -1 from $w\tilde w$) and six little group (three $SU(2)_L$ and three $SU(2)_R$) indices. Thus, we can write down the 3-point amplitude as

$$
A(1_{a\dot a}2_{b\dot b}3_{c\dot c}) = i\Gamma_{abc}\Gamma_{\dot a\dot b\dot c}. \qquad (3.30)
$$

3.4 One gluon two fermion amplitude

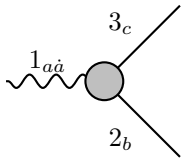


Figure 2: 3-point Feynman diagram for two fermions and one gluon. Vector particles are the wavy lines and the fermions are the straight lines.

From Fig. 2, we can see that the amplitude has four (three $SU(2)_L$ and one $SU(2)_R$) little group indices. Using the objects previously defined, we can construct the amplitude with Γ_{abc} that has three little group indices and mass dimensions 1/2 and $\tilde{u}_{\dot{a}}$ that has one little group index and mass dimensions 1/2. Thus, we get the amplitude as,

$$A(1_{a\dot{a}}2_b3_c) = i\Omega_{f_2f_3}\Gamma_{abc}\tilde{u}_{1\dot{a}} \tag{3.31}$$

where the $\Omega_{f_2f_3}$ is the $USp(2N_f)$ symplectic form introduced to keep track of the flavor.

4 BCFW recursion relation in 6D

In this section, we will review the on-shell recursion relation [1, 11] that will help us construct higher-point amplitudes using the 3-point amplitude that we constructed in the previous section.

Consider a n-point amplitude A_n that has $n-$particles with momenta p_i. We begin the construction with some shifted momenta

$$\hat{p}_i = p_i + zr_i^\mu; \quad z \in \mathbb{C}, \tag{4.1}$$

such that the new momenta obey the relations

$$\sum_i z_i^\mu = 0, \tag{4.2a}$$

$$r_i \cdot r_j = 0, \tag{4.2b}$$

$$p_i \cdot r_i = 0. \tag{4.2c}$$

From Eq.(4.2a), we note that the momentum $\sum_i \hat{p}_i$ is conserved, and from Eq.(4.2b) and Eq. (4.2c), we can note that all shifted momenta are on-shell $\hat{p}_i^2 = 0$.

Using these shifted momenta, we can construct a shifted amplitude as a function of z, $\hat{A}_n(z)$. Notice that by setting $z = 0$, we get back the unshifted amplitude A_n. Now we define internal propagators with $P_I = \sum_{i\in I} p_i$, such that $P_I^2 \neq 0$. Then, in the shifted diagram, we have

$$\hat{P}_I^2 = P_I^2 + z2P_I \cdot R_I, \;\; \hat{P}_I = \sum_{i\in I}\hat{p}_i, \;\; R_I = \sum_{i\in I} r_i,$$
$$\hat{P}_I^2 = -\frac{P_I^2}{z_I}(z - z_I), \;\; z_I = \frac{-P_I^2}{2P_I \cdot R_I}. \tag{4.3}$$

In the case where A_n is a tree-level amplitude, $\hat{A}_n(z)$ has only simple poles from the shifted propagators $\hat{P}_I^{-2}$. Consider the function $z^{-1}\hat{A}_n(z)$ in the complex plane. Pick a contour that surrounds the origin. Then, by the Cauchy residue theorem,

$$\oint_{C_0} \frac{\hat{A}_n(z)}{z} dz = A_n, \tag{4.4}$$

where C_0 is a contour of radius $\epsilon << 1$ around 0. Deforming the contour around the other simple poles, we get,

$$\begin{aligned} A_n &= \oint_{C_\infty} \frac{\hat{A}_n(z)}{z} dz - \sum_{z_I} \oint_{C_{z_I}} \frac{\hat{A}_n(z)}{z} \\ &= -\sum_{z_I} \text{Res}_{z=z_I} \frac{\hat{A}_n(z)}{z} + B_n \end{aligned} \tag{4.5}$$

where B_n is the residue at $z = \infty$. For the rest of this discussion, we assume $B_n = 0$. In the limit $z = z_I$, the propagator $\hat{P}_I$ goes on-shell and the amplitude factorizes into two sub-amplitudes [22, 23],

$$\text{Res}_{z=z_I} \frac{\hat{A}_n(z)}{z} = -\hat{A}_L(z_I) \frac{1}{P_I^2} \hat{A}_R(z_I). \tag{4.6}$$

Thus, from this construction, the n-point amplitude can be constructed as a sum of all possible factorization diagrams of amplitudes with fewer than n points.

The Britto-Cachazo-Feng-Witten (BCFW) recursion relation [11] is one such recursion, where we shift the momenta of two particles (say p_1, p_2), and set r_i for the rest to 0. In 6D, we can write the shifted momenta as

$$\hat{p}_1^{AB} = p_1^{AB} + z r^{AB}; \quad \hat{p}_2^{AB} = p_2^{AB} - z r^{AB}. \tag{4.7}$$

For the shifted momenta to satisfy the conditions in Eq.(4.2b), Eq.(4.2c), a standard choice for r is to set it as the linear combination of the polarization vector of p_1, with the gauge choice $q = p_2$, then,

$$r^{AB} = x^a \tilde{x}^{\dot{a}} \varepsilon^{AB}_{1a\dot{a}}, \ \varepsilon^{AB}_{1a\dot{a}} = \frac{[A|1_a\rangle[1_{\dot{a}}|2^b\rangle\langle 2_b|^{B]}}{s_{12}}. \tag{4.8}$$

Similarly,

$$r_{AB} = x^a \tilde{x}^{\dot{a}} \varepsilon_{1ABa\dot{a}}, \ \varepsilon_{1ABa\dot{a}} = \frac{[_A|1_{\dot{a}}]\langle 1_a|2^{\dot{b}}][2_{\dot{b}}|_{B]}}{s_{12}}. \tag{4.9}$$

In spinor-helicity language, these momentum shifts can be represented as

$$|\hat{1}_a\rangle^A = |1_a\rangle^A + z x_a \frac{[\mathbf{1}|2^b\rangle}{s_{12}} |2_b\rangle^A, \tag{4.10a}$$

$$|\hat{2}_b\rangle^A = |2_b\rangle^A + z \frac{[\mathbf{1}|2_b\rangle}{s_{12}} |\mathbf{1}\rangle^A, \tag{4.10b}$$

$$|\hat{1}_{\dot{a}}]_A = |1_{\dot{a}}]_A + z \tilde{x}_{\dot{a}} \frac{\langle \mathbf{1}|2^{\dot{b}}]}{s_{12}} |2_{\dot{b}}]_A, \tag{4.10c}$$

$$|\hat{2}_{\dot{b}}]_A = |2_{\dot{b}}]_A + z\frac{\langle \mathbf{1}|2_{\dot{b}}]}{s_{12}}|\mathbf{1}]_A \tag{4.10d}$$

where we have introduced the notation $|\mathbf{1}\rangle = x^a\,|1_a\rangle$ and $|\mathbf{1}] = \tilde{x}^{\dot{a}}|1_{\dot{a}}]$.

From Eq.(4.8), one can notice that the polarization for the shift momentum r_{AB} is an arbitrary linear combination of polarizations of the particle 1 parametrized by $x_a\tilde{x}_{\dot{a}}$, which enter on the RHS. Using the recursion relation, one also gets an amplitude with the particle 1's polarization parametrized by the x's. Finally, the recursion relation in 6D with k as the propagator momentum is expressed as

$$x^a\tilde{x}^{\dot{a}}A(1_{a\dot{a}}, 2_{b\dot{b}}, \ldots) = \sum_{\text{all factorizations}} \left(\frac{-i}{k^2}\right) x^a\tilde{x}^{\dot{a}}A(\hat{1}_{a\dot{a}}, \ldots, \hat{k}_{c\dot{c}})A(\hat{2}_{b\dot{b}}, \ldots, \hat{k}^{c\dot{c}}). \tag{4.11}$$

4.1 Four gluon amplitude

Before applying the recursion relation, we can first try to construct the amplitudes from the little group covariant building blocks. The 4-point amplitude has mass-dimension 0 and four $SU(2)_L$ and four $SU(2)_R$ little group indices (as shown in Fig. 3). For this, one can construct two possible ansatzes

$$\begin{aligned} A(1_{a\dot{a}}2_{b\dot{b}}3_{c\dot{c}}4_{d\dot{d}}) =& \frac{1}{D}\langle 1_a 2_b 3_c 4_d\rangle[1_{\dot{a}}2_{\dot{b}}3_{\dot{c}}4_{\dot{d}}], \\ A(1_{a\dot{a}}2_{b\dot{b}}3_{c\dot{c}}4_{d\dot{d}}) =& \frac{1}{D}\langle 1_a|2_{\dot{b}}]\langle 2_b|3_{\dot{c}}]\langle 3_c|4_{\dot{d}}]\langle 4_a|1_{\dot{a}}] + \text{perms}. \end{aligned} \tag{4.12}$$

Here, the denominator D needs to have mass dimensions 4. From the numerators, we can notice that

$$\begin{aligned} &\epsilon_{ABCD}\lambda^A{}_{1a}\lambda^B{}_{2b}\lambda^C{}_{3c}\lambda^D{}_{4d}\epsilon^{A'B'C'D'}\tilde{\lambda}_{A'1\dot{a}}\tilde{\lambda}_{B'2\dot{b}}\tilde{\lambda}_{C'3\dot{c}}\tilde{\lambda}_{D'4\dot{d}} \\ =&(\delta^{B'}_A\delta^{C'}_B\delta^{D'}_C\delta^{A'}_D + \text{perms})\lambda^A{}_{1a}\lambda^B{}_{2b}\lambda^C{}_{3c}\lambda^D{}_{4d}\tilde{\lambda}_{A'1\dot{a}}\tilde{\lambda}_{B'2\dot{b}}\tilde{\lambda}_{C'3\dot{c}}\tilde{\lambda}_{D'4\dot{d}} \\ =&\langle 1_a|2_{\dot{b}}]\langle 2_b|3_{\dot{c}}]\langle 3_c|4_{\dot{d}}]\langle 4_a|1_{\dot{a}}] + \text{perms}. \end{aligned} \tag{4.13}$$

Thus we note that both the ansatzes are equivalent. Now we have to fix the denominator. Unlike the 3-point situation, here we have non-vanishing Mandelstam variables that can be part of the denominator. For the color order $(1,2,3,4)$, there are two independent Mandelstam variables, s_{12}, s_{14}. Since D has mass dimensions 4 and each Mandelstam variable has mass dimensions 2, we can guess the amplitude to be of the form

$$A(1_{a\dot{a}}2_{b\dot{b}}3_{c\dot{c}}4_{d\dot{d}}) = -\frac{i}{s_{12}s_{14}}\langle 1_a 2_b 3_c 4_d\rangle[1_{\dot{a}}2_{\dot{b}}3_{\dot{c}}4_{\dot{d}}]. \tag{4.14}$$

We can now verify that our choice for the poles is correct using the recursion relation.

From recursion

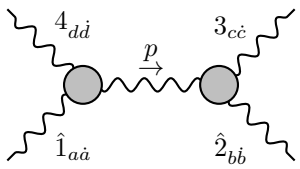


Figure 3: Feynman diagram for four gluons with the color ordering (1,2,3,4). The hats(ˆ) denote the shifted legs.

For the 4-point amplitude with the legs $1,2$ shifted, the BCFW recursion relation Eq.(4.11) is given as

$$\begin{aligned} x^a\tilde{x}^{\dot{a}}A(1_{a\dot{a}}2_{b\dot{b}}3_{c\dot{c}}4_{d\dot{d}}) =& -\frac{i}{s_{14}}x^a\tilde{x}^{\dot{a}}A_3(1_{a\dot{a}}4_{d\dot{d}}p_{e\dot{e}})A_3(2_{b\dot{b}}(-p)^{e\dot{e}}3_{c\dot{c}}) \\ =& \frac{i}{s_{14}}x^a\tilde{x}^{\dot{a}}\Gamma_{ade}\Gamma_{\dot{a}d\dot{e}}\Gamma_b{}^e{}_c\Gamma_{\dot{b}}{}^{\dot{e}}{}_{\dot{c}}. \end{aligned} \tag{4.15}$$

The tensorial contractions simplify as (detailed calculation can be found in appendix B)

$$\Gamma_{ade}\Gamma_b{}^e{}_c = \frac{1}{\tilde{u}_{p\dot{e}}\tilde{u}^{\dot{e}}_{-p}}\langle\hat{1}_a\hat{2}_b3_c4_d\rangle, \tag{4.16a}$$

$$\Gamma_{\dot{a}d\dot{e}}\Gamma_{\dot{b}}{}^{\dot{e}}{}_{\dot{c}} = \frac{1}{u_{pe}u^e_{-p}}[\hat{1}_{\dot{a}}\hat{2}_{\dot{b}}3_{\dot{c}}4_{\dot{d}}], \tag{4.16b}$$

$$\frac{1}{\tilde{u}_{p\dot{e}}\tilde{u}^{\dot{e}}_{-p}}\frac{1}{u_{pe}u^e_{-p}} = -\frac{1}{s_{12}}. \tag{4.16c}$$

Then, the 4-point amplitude is,

$$x^a\tilde{x}^{\dot{a}}A(1_{a\dot{a}}2_{b\dot{b}}3_{c\dot{c}}4_{d\dot{d}}) = -\,x^a\tilde{x}^{\dot{a}}\frac{i}{s_{12}s_{14}}\langle\hat{1}_a\hat{2}_b3_c4_d\rangle[\hat{1}_{\dot{a}}\hat{2}_{\dot{b}}3_{\dot{c}}4_{\dot{d}}].$$

The final step is to remove the z dependence on the shifted spinors; to that end, we note that

$$|\mathbf{1}\rangle = x^a\,|\hat{1}_a\rangle = x^a(|1_a\rangle + z\tilde{x}^{\dot{a}}x_a\frac{[1_{\dot{a}}|2^b\rangle}{s_{12}}\,|2_b\rangle) = x^a\,|1_a\rangle\,, \tag{4.17a}$$

$$\langle\mathbf{1}\hat{2}_b3_c4_d\rangle = \langle\mathbf{1}2_b3_c4_d\rangle + z\tilde{x}^{\dot{a}}\frac{[1_{\dot{a}}|2_b\rangle}{s_{12}}\langle\mathbf{1}13_c4_d\rangle = x^a\langle1_a2_b3_c4_d\rangle. \tag{4.17b}$$

Using these,

$$\begin{aligned} & x^a\tilde{x}^{\dot{a}}A(1_{a\dot{a}}2_{b\dot{b}}3_{c\dot{c}}4_{d\dot{d}}) = -\,x^a\tilde{x}^{\dot{a}}\frac{i}{s_{12}s_{14}}\langle1_a2_b3_c4_d\rangle[1_{\dot{a}}2_{\dot{b}}3_{\dot{c}}4_{\dot{d}}] \\ \implies & A(1_{a\dot{a}}2_{b\dot{b}}3_{c\dot{c}}4_{d\dot{d}}) = -\frac{i}{s_{12}s_{14}}\langle1_a2_b3_c4_d\rangle[1_{\dot{a}}2_{\dot{b}}3_{\dot{c}}4_{\dot{d}}]. \end{aligned} \tag{4.18}$$

4.2 Four fermion amplitude

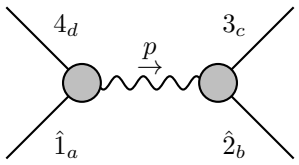


Figure 4

Starting from the BCFW recursion relation Eq.(4.11),

$$x^a\tilde{x}^{\dot{a}}A(1_a2_b3_c4_d) = -\frac{i}{s_{14}}x^a\tilde{x}^{\dot{a}}A_3(4_d\hat{1}_ap_{e\dot{e}})A_3(\hat{2}_b3_c(-p)^{e\dot{e}})$$

$$= \frac{i}{s_{14}} x^a \tilde{x}^{\dot{a}} \Omega_{f_4 f_1} \Omega_{f_2 f_3} \Gamma_{ade} \tilde{u}_{p\dot{e}} \Gamma_{bc}{}^{e} \tilde{u}^{\dot{e}}_{-p}. \tag{4.19}$$

Using Eq. (4.16),

$$\begin{aligned} x^a \tilde{x}^{\dot{a}} A(1_a 2_b 3_c 4_d) =& \frac{i}{s_{14}} x^a \tilde{x}^{\dot{a}} \Omega_{f_4 f_1} \Omega_{f_2 f_3} \frac{1}{\tilde{u}_{p\dot{e}} \tilde{u}^{\dot{e}}_{-p}} \langle \hat{1}_a \hat{2}_b 3_c 4_d \rangle \tilde{u}_{p\dot{e}} \tilde{u}^{\dot{e}}_{-p} \\ \implies A(1_a 2_b 3_c 4_d) =& \frac{i}{s_{14}} \Omega_{f_4 f_1} \Omega_{f_2 f_3} \langle 1_a 2_b 3_c 4_d \rangle. \end{aligned}$$

However, for the color-ordered four fermion amplitude, there is another possible flavor branch $\Omega_{f_1 f_2} \Omega_{f_3 f_4}$. Choosing $1, 2$ as the shifted legs, it is not possible to get the contribution of this branch using BCFW. Since this flavor branch can also be seen as a rotation of the $\Omega_{f_4 f_1} \Omega_{f_2 f_3}$ branch. Thus, we get the total amplitude as

$$A(1_a 2_b 3_c 4_d) = \frac{i}{s_{14}} \Omega_{f_4 f_1} \Omega_{f_2 f_3} \langle 1_a 2_b 3_c 4_d \rangle + \frac{i}{s_{12}} \Omega_{f_1 f_2} \Omega_{f_3 f_4} \langle 1_a 2_b 3_c 4_d \rangle \tag{4.20}$$

4.3 Two gluon two fermion amplitude

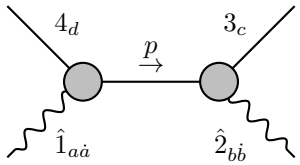


Figure 5

Starting from the BCFW recursion relation Eq. (4.11),

$$\begin{aligned} x^a \tilde{x}^{\dot{a}} A(1_{a\dot{a}} 2_{b\dot{b}} 3_c 4_d) =& -\frac{i}{s_{14}} x^a \tilde{x}^{\dot{a}} A_3(\hat{1}_{a\dot{a}} p_e 4_d) A_3(\hat{2}_{b\dot{b}} 3_c (-p)^e) \\ =& \frac{i}{s_{14}} x^a \tilde{x}^{\dot{a}} \Gamma_{ade} \tilde{u}_{1\dot{a}} \Gamma_{bc}{}^{e} \tilde{u}_{2\dot{b}}. \end{aligned} \tag{4.21}$$

Using Eq. (4.16),

$$\begin{aligned} x^a \tilde{x}^{\dot{a}} A(1_{a\dot{a}} 2_{b\dot{b}} 3_c 4_d) =& \frac{i}{s_{14}} x^a \tilde{x}^{\dot{a}} \Omega_{f_3 f_4} \frac{1}{\tilde{u}_{p\dot{e}} \tilde{u}^{\dot{e}}_{-p}} \langle \hat{1}_a \hat{2}_b 3_c 4_d \rangle \tilde{u}_{1\dot{a}} \tilde{u}_{2\dot{b}} \\ =& -\frac{i}{s_{12} s_{14}} x^a \tilde{x}^{\dot{a}} \Omega_{f_3 f_4} \langle \hat{1}_a \hat{2}_b 3_c 4_d \rangle u_{pe} u^{e}_{-p} \tilde{u}_{1\dot{a}} \tilde{u}_{2\dot{b}} \\ =& \frac{1}{s_{12} s_{14}} x^a \tilde{x}^{\dot{a}} \Omega_{f_3 f_4} \langle \hat{1}_a \hat{2}_b 3_c 4_d \rangle [\hat{1}_{\dot{a}} | \not{p}_4 | \hat{2}_{\dot{b}}] \\ \implies A(1_{a\dot{a}} 2_{b\dot{b}} 3_c 4_d) =& \frac{1}{s_{12} s_{14}} \Omega_{f_3 f_4} \langle 1_a 2_b 3_c 4_d \rangle [1_{\dot{a}} | \not{p}_4 | 2_{\dot{b}}]. \end{aligned} \tag{4.22}$$

5 Supersymmetry in 6D

Symmetries are an essential property of a field theory that help simplify calculations of scattering amplitudes. In addition to the gauge symmetry and flavor symmetry mentioned in section 2,

this theory is also invariant under the Lorentz group $SO(1,5)$. The elements of the group are generated by the momentum (P^μ) and rotation ($M^{\mu\nu}$) generators called the Poincaré generators that satisfy

$$[P^\mu, P^\nu] = 0, \tag{5.1a}$$
$$[M^{\mu\nu}, P^\rho] = i(g^{\nu\rho}P^\mu - g^{\mu\rho}P^\nu), \tag{5.1b}$$
$$[M^{\mu\nu}, M^{\rho\sigma}] = i(g^{\nu\rho}M^{\mu\sigma} - g^{\mu\rho}M^{\nu\sigma} - g^{\nu\sigma}M^{\mu\rho} + g^{\mu\sigma}M^{\rho\nu}). \tag{5.1c}$$

In addition to the usual Poincaré generators, we can extend the symmetry by introducing fermionic generators that change the spin of the particles. These fermionic (supersymmetric) generators satisfy the anti-commutation relation, depending on the reality conditions that can be imposed on the irreducible representation of the spinors.

$$\{Q^{AI}, Q^{BJ}\} = P^\mu \Gamma_\mu^{AB} \Omega^{IJ} \tag{5.2}$$

where Γ_μ are the 6D gamma matrices that obey the Clifford algebra, I, J are used to label sets of supersymmetry generators, P^μ is the momentum operator, and Ω^{IJ} is a symplectic form corresponding to internal symmetry of fermionic generators.

In the usual 4D case, the femionic generators satisfy

$$\{Q^{AI}, \bar{Q}^B_J\} = P^\mu \gamma_\mu^{AB} \delta^I_J. \tag{5.3}$$

Since we have complex Weyl spinors as the irreducible representations. Here, q's and $\bar{q}$'s are independent of each other and obey the anti-commutation relation Eq. (5.3). The automorphism group that preserves this relation needs to satisfy

$$U^{\dagger I}{}_J \delta^J_K U^K_L = \delta^I{}_L. \tag{5.4}$$

Thus, this internal symmetry (known as R-symmetry) of the generators is the $U(\mathcal{N})$ group, where the $\mathcal{N}$ corresponds to the number of sets of supersymmetric generators. The maximal supersymmetric system with spin ≤ 1 is given by $\mathcal{N} = 4$ with $4 \times 2 = 8$ complex DoF $\implies$ 16 generators.

In the 6D case, the irreducible representation of the spinors satisfies a symplectic-Majorana Weyl (SMW) condition, such that

$$\psi^{Aa} = \Omega^{AB} \mathcal{C}^a{}_b (\psi^b_B)^* \tag{5.5}$$

where $\mathcal{C}$ is the charge conjugation matrix. Also, since the charge-conjugated spinor has the same chirality, we can impose the Majorana conditions in the chiral and anti-chiral representations separately. This results in anti-commutation relations for the chiral and anti-chiral representations simultaneously as,

$$\{Q^{AI}, Q^{BJ}\} = P^\mu \bar{\Sigma}_\mu^{AB} \Omega^{IJ}, \tag{5.6a}$$
$$\{\tilde{Q}_{AI}, \tilde{Q}_{BJ}\} = P^\mu \Sigma_{\mu\, AB} \Omega_{IJ}. \tag{5.6b}$$

Thus, the automorphism group for these supersymmetric generators needs to satisfy the symplectic condition,

$$U^I{}_J \Omega^{JK} U^{\dagger L}{}_K = \Omega^{IL} \tag{5.7}$$

which gives rise to the R-symmetry $USp(2\mathcal{N}_+) \times USp(2\mathcal{N}_-)$, where $\mathcal{N}_+, \mathcal{N}_-$ correspond to the number of sets of chiral and anti-chiral generators, respectively.

In 6D, the minimal theory when considering supersymmetry is the $\mathcal{N} = (1,0)$ (or equivalently $\mathcal{N} = (0,1)$) gauge theory [24]. Considering non-gravity theories, the maximally supersymmetric case in 6D is given by the $\mathcal{N} = (1,1)$ [14] and $\mathcal{N} = (2,0)$ [25] theories. In the subsequent sections, we will introduce these theories and their particle spectrum in more detail.

With the inclusion of gravity interaction, the minimal supersymmetric theory is the $\mathcal{N} = (1,0)$ supergravity, which contains a graviton multiplet in addition to the SQCD multiplets [26]. The half-maximal supergravity theories are the $\mathcal{N} = (1,1)$ non-chiral and the $\mathcal{N} = (2,0)$ chiral supergravities [27]. We will study these two theories as the double copy of $\mathcal{N} = (1,0)$ SQCD theory in section 7.

In addition to this, there also exists supergravity theory with 24 supercharges, $\mathcal{N} = (2,1)$ supergravity, that is studied in [28]. Finally, there are three inequivalent maximal supergravity theories $\mathcal{N} = (2,2)$, $\mathcal{N} = (3,1)$ and $\mathcal{N} = (4,0)$. The $\mathcal{N} = (2,2)$ supergravity is a well-defined non-chiral theory that has a Lagrangian description [29]. However, the chiral theories ($\mathcal{N} = (3,1)$ and $\mathcal{N} = (4,0)$) are classified as exotic supergravity theories due to the lack of a traditional graviton multiplet [30]. A description of the field content in these theories can be found in [31].

5.1 $\mathcal{N} = (1,1)$ SYM

For the $\mathcal{N} = (1,1)$ theory, we have the algebra for the chiral and anti-chiral components as

$$\{q^{A\alpha}, q^{B\beta}\} = p^{AB}\epsilon^{\alpha\beta}, \tag{5.8a}$$
$$\{\tilde{q}_{A\dot{\alpha}}, \tilde{q}_{B\dot{\beta}}\} = p_{AB}\epsilon_{\dot{\alpha}\dot{\beta}}, \tag{5.8b}$$

where $q^{A\alpha}$ is the supermomentum corresponding to a single particle and α, β are the R-symmetry indices of $SU(2) \times SU(2)$. The supercharges for the chiral (anti-chiral) algebras can be expressed as products of chiral (anti-chiral) spinors and Grassmann variables,

$$q^{A\alpha} = \lambda_a^A \eta^{a\alpha}; \qquad \tilde{q}_{A\alpha} = \tilde{\lambda}_{Aa}\tilde{\eta}_\alpha^a, \tag{5.9}$$

here the η's and $\tilde{\eta}$'s are complex. Unlike the 4D case, we need both the chiral and anti-chiral variables to describe the theory. This naively looks like we have double the supercharges. However, these η's and $\tilde{\eta}$'s are self-conjugate and only half the components are independent [14].

To describe only the physical components, we truncate the R-symmetry index with fixed two-component vectors, effectively reducing the supercharges by half. The truncated supercharges are now given by dropping the R-symmetry index

$$q^A = \lambda_a^A \eta^a; \qquad \tilde{q}_A = \tilde{\lambda}_{Aa}\tilde{\eta}^a. \tag{5.10}$$

In this superspace, we can write down all the possible fields into a vector superfield as,

$$\mathcal{V}^{(1,1)} = \phi^1 + \eta_a\chi^a + \tilde{\eta}^{\dot{a}}\tilde{\chi}_{\dot{a}} + \eta_a\tilde{\eta}^{\dot{a}}g^a{}_{\dot{a}} + \eta^2\phi^2 + \tilde{\eta}^2\phi^3 + \eta^2\tilde{\eta}^{\dot{a}}\bar{\tilde{\chi}}_{\dot{a}} + \tilde{\eta}^2\eta_a\bar{\chi}^a + \eta^2\tilde{\eta}^2\phi^4 \tag{5.11}$$

where, ϕ's are 4 scalar fields, $\chi, \tilde{\chi}$'s are fermionic fields and g is the gluon field.

5.2 $\mathcal{N} = (1,0)$ SQCD

Contrary to the $\mathcal{N} = (1,1)$ theory, the superalgebra is extended to only the chiral part (or anti-chiral for the $\mathcal{N} = (0,1)$ theory), giving rise to just half of the supercharges

$$q^A = \lambda_a^A \eta^a. \tag{5.12}$$

There are three possible massless superfields in this theory, a vector multiplet, a hyper multiplet and a tensor multiplet:

$$\mathcal{V}_{\dot{a}}^{(1,0)} = \lambda_{\dot{a}} + \eta_a g^a_{\dot{a}} + \eta^2 \bar{\lambda}_{\dot{a}}, \tag{5.13a}$$

$$\Phi^{(1,0)} = (\phi + \eta_a \psi^a + \eta^2 \phi') \text{ and } (\eta^2 \bar{\phi} + \eta_a \bar{\psi}^a + \bar{\phi}'), \tag{5.13b}$$

$$\mathcal{T}_a^{(1,0)} = \chi_a + \eta^b \big(B_{ab} + \frac{\epsilon_{ab}}{\sqrt{2}}\varphi\big) + \eta^2 \bar{\chi}_a \tag{5.13c}$$

where the vector multiplet contains a gluon field and two fermionic fields, the hypermultiplet contains two scalar fields and a fermionic field (with a CPT conjugate), and the tensor multiplet contains two fermionic fields, a scalar field, and a self-dual B-field.

Connecting back to the discussion of the Lagrangian from section 2, the 6D SQCD theory has a vector multiplet and N_f copies of the hypermultiplet. These N_f copies give rise to the flavor symmetry $USp(2N_f)$.

5.3 $\mathcal{N} = (2,0)$

The chiral $\mathcal{N} = (2,0)$ theory has 16 supercharges of the same chirality, and has the $USp(4)$ R-symmetry satisfying the algebra

$$\{q^{AI}, q^{BJ}\} = p^{AB}\Omega^{IJ} \tag{5.14}$$

where Ω^{IJ} is the $USp(4)$ symplectic form. Similar to the previous cases, the η^I's are self-conjugate, and we need to truncate half the supercharges. This superspace can be described using the variables $(\lambda, \eta^1, \eta^2)$.

The only multiplet in this theory is a tensor multiplet, which in the harmonic superspace can be written as

$$\begin{aligned}\mathcal{T}^{(2,0)} =& \phi_1 + \psi_1^a \eta_a^1 + \psi_2^a \eta_a^2 + (\eta_1)^2 \phi_2 + (\eta_2)^2 \phi_3 + (\eta_1.\eta_2)\phi_4 \\ &+ \eta^{1a}\eta^{2b} B_{(ab)} + (\eta^1)^2 \eta^{2a} \tilde{\psi}_{2a} + (\eta^2)^2 \eta^{1a} \tilde{\psi}_{1a} + (\eta^1)^2 (\eta^2)^2 \phi_5\end{aligned} \tag{5.15}$$

where ϕ's are five scalar fields, B_{ab} is a self-dual field and $\psi, \tilde{\psi}$'s are fermionic fields.

5.4 Super-BCFW

The discussion for the super-BCFW relations builds upon the conditions for the BCFW shifts. In addition to conserving the momentum, we also need to conserve the supermomentum, $Q(\tilde{Q})$. This is accomplished be shifting the Grassmann variables. In the $\mathcal{N} = (1,1)$ theory, the shifts are given as [14]

$$\eta_{\hat{1}a} = \eta_{1a} + z\frac{x_a[\mathbf{1}|2^b\rangle \eta_{2b}}{s_{12}}, \tag{5.16a}$$

$$\eta_{\hat{2}a} = \eta_{2a} + z\frac{x^b[\mathbf{1}|2_a\rangle\eta_{1b}}{s_{12}}, \tag{5.16b}$$

$$\tilde{\eta}_{\hat{1}\dot{a}} = \tilde{\eta}_{1\dot{a}} + z\frac{\tilde{x}_{\dot{a}}\langle\mathbf{1}|2^{\dot{b}}]\tilde{\eta}_{2\dot{b}}}{s_{12}}, \tag{5.16c}$$

$$\tilde{\eta}_{\hat{2}\dot{a}} = \tilde{\eta}_{2\dot{a}} + z\frac{\tilde{x}^{\dot{b}}\langle\mathbf{1}|2_{\dot{a}}]\tilde{\eta}_{1\dot{b}}}{s_{12}}. \tag{5.16d}$$

For the $\mathcal{N} = (1,0)$ theory, we can consider only the shifts of the chiral Grassmann variables to conserve the supermomentum $\sum_i q_i$.

Finally, to stitch the left part and right part of the superamplitudes, we need to sum over the helicity states of the propagator. This is done by integrating the two smaller multiplicity amplitudes as shown:

$$\mathcal{A}_n(1,2,\ldots,n) = \int \frac{1}{2}d\eta_{\hat{p}a}d\eta^a_{\hat{p}} \int \frac{1}{2}d\tilde{\eta}_{\hat{p}\dot{a}}d\tilde{\eta}^{\dot{a}}_{\hat{p}}\mathcal{A}_L(\hat{1},2,\ldots,i,\hat{p})\frac{i}{p}\mathcal{A}_R(-\hat{p},i+1,\ldots,\hat{n}) \tag{5.17}$$

where p is the propagator momentum.

In the next sub-sections, we will discuss the 3-point amplitudes in $\mathcal{N} = (1,1)$ and $\mathcal{N} = (1,0)$ theories, and use this recursion relation to construct 4-point amplitudes.

5.5 3-point superamplitudes

SYM Theory

The three vector amplitude in $\mathcal{N} = (1,1)$ theory is given as [14]

$$\mathcal{A}_3^{(1,1)}(\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3) = \frac{i}{2}[\delta(\sum_i q_i^A)\delta(\sum_i \tilde{q}_{iA})]^2\delta(\sum_i w_i^b\eta_b)\delta(\sum_i \tilde{w}_i^{\dot{b}}\eta_{\dot{b}}). \tag{5.18}$$

Before studying the superamplitudes in the $\mathcal{N} = (1,0)$ theory, it would be fruitful to rewrite the above amplitude in a form where the chiral and anti-chiral parts are separable. To this end, taking inspiration from the representation of 3-point superamplitudes in 5D theories from [7], we introduce two reference spinors $|\rho\rangle^A$ and $|\tilde{\rho}]_A$ that satisfy the following properties

$$\tilde{w}_{i\dot{a}} = \langle\rho|i_{\dot{a}}], \tag{5.19a}$$

$$w_{ia} = [\tilde{\rho}|i_{\dot{a}}\rangle. \tag{5.19b}$$

Then, the three gluon amplitude Eq.(3.30) can be rewritten as

$$A(1_{a\dot{a}}2_{b\dot{b}}3_{c\dot{c}}) = i\langle 1_a 2_b 3_c\rho\rangle[1_{\dot{a}}2_{\dot{b}}3_{\dot{c}}\tilde{\rho}]. \tag{5.20}$$

Superamplitudes ($\mathcal{A}$) are polynomials in the Grassmann variables (η) that act as a bookkeeping function for the scattering amplitudes of the component fields in a multiplet [32]. Thus, the three gluon amplitude Eq.(5.20) is also a component of the three vector superamplitude. Using the vector superfield Eq.(5.11), we have

$$\prod_{i=1}^{3}\frac{\partial}{\partial\eta_i^a}\frac{\partial}{\partial\tilde{\eta}_i^{\dot{a}}}\mathcal{A}_3^{(1,1)}(\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3) = i\langle 1_a 2_b 3_c\rho\rangle[1_{\dot{a}}2_{\dot{b}}3_{\dot{c}}\tilde{\rho}]. \tag{5.21}$$

Also, all superamplitudes have supermomentum conservation in the form of a delta function as a prefactor. In 6D, this supermomentum delta function takes the form

$$\delta^{(4)}(Q) = \frac{1}{4!}\langle QQQQ\rangle \tag{5.22}$$

where $Q = \sum_i q_i$ and $\tilde{Q} = \sum_i \tilde{q}_i$.

However, for the 3-point superamplitude, we only have three independent Q's. So instead of the fourth Q, we can use the reference vector ρ (Eq.(5.19a)) to take the fourth spot. From this and Eq.(5.21), one can express the three vector states amplitude (Eq.(5.18)) in another way as

$$\mathcal{A}_3^{(1,1)}(\mathcal{V}_1, \mathcal{V}_2, \mathcal{V}_3) = \frac{i}{(3!)^2}\langle QQQ\rho\rangle[\tilde{Q}\tilde{Q}\tilde{Q}\tilde{\rho}]. \tag{5.23}$$

SQCD Theory

In the SQCD theory, the external states can belong to either the vector multiplet or the hyper multiplet. The possible 3-point interactions are either three vectors as external states, or two hypers and one vector as external states.

The three gluon amplitude Eq.(3.30) will be a component amplitude of the three vector superamplitude. Hence, the superamplitude can be represented similarly to the $\mathcal{N} = (1,1)$ theory with the $\tilde{\eta}$'s projected out. Thus, the amplitude can be written as

$$\mathcal{A}^{(1,0)}(\mathcal{V}_{1\dot{a}}, \mathcal{V}_{2\dot{b}}, \mathcal{V}_{3\dot{c}}) = \frac{i}{3!}\langle QQQ\rho\rangle[1_{\dot{a}}2_{\dot{b}}3_{\dot{c}}\tilde{\rho}]. \tag{5.24}$$

Similarly, for the two hypers and one vector superamplitude, we have the expression for a component amplitude of two fermion one gluon Eq.(3.31). From this expression, we can project the gamma tensor similar to Eq.(5.24) to get the amplitude as

$$\mathcal{A}^{(1,0)}(\mathcal{V}_{1\dot{a}}, \Phi_2, \Phi_3) = \frac{i}{3!}\Omega_{f_2 f_3}\langle QQQ\rho\rangle\tilde{u}_{1\dot{a}}. \tag{5.25}$$

5.6 4-point superamplitudes

The 4-point superamplitudes in SYM are derived in [8] and the superamplitudes in SQCD are derived in [20]. In this section, we give a brief description of the expressions.

SYM Theory

To construct the four vector amplitude in the SYM theory, we can start by reviewing the 4 gluon amplitude Eq.(4.18). Since the four gluon amplitude is a component of the superamplitude, we know that when differentiating with the proper η's, one should get back Eq.(4.18), i.e.,

$$\prod_{i=1}^{4}\frac{\partial}{\partial\eta_i^a}\frac{\partial}{\partial\tilde{\eta}_i^{\dot{a}}}\mathcal{A}^{(1,1)}(\mathcal{V}_1, \mathcal{V}_2, \mathcal{V}_3, \mathcal{V}_4) = -\frac{i}{s_{12}s_{14}}\langle 1_a 2_b 3_c 4_d\rangle[1_{\dot{a}}2_{\dot{b}}3_{\dot{c}}4_{\dot{d}}]. \tag{5.26}$$

From this equation, one can guess the superamplitude to be

$$\mathcal{A}^{(1,1)}(\mathcal{V}_1, \mathcal{V}_2, \mathcal{V}_3, \mathcal{V}_4) = -\frac{i}{s_{12}s_{14}}\frac{1}{(4!)^2}\langle QQQQ\rangle[\tilde{Q}\tilde{Q}\tilde{Q}\tilde{Q}] = -\frac{i}{s_{12}s_{14}}\delta^{(4)}(Q)\delta^{(4)}(\tilde{Q}). \tag{5.27}$$

SQCD Theory

Similar to the three vector superamplitude Eq.(5.25), the superamplitude with four vectors as external states can be written by projecting out the $\tilde{\eta}$'s from the SYM amplitude Eq.(5.27). Thus,

$$\mathcal{A}^{(1,0)}(\mathcal{V}_{1\dot{a}},\mathcal{V}_{2\dot{b}},\mathcal{V}_{3\dot{c}},\mathcal{V}_{4\dot{d}}) = -\frac{i}{s_{12}s_{14}}\delta^{(4)}(Q)[1_{\dot{a}}2_{\dot{b}}3_{\dot{c}}4_{\dot{d}}]. \tag{5.28}$$

Similarly, the four hyper superamplitudes and the two hyper two vector superamplitudes are given as

$$\mathcal{A}^{(1,0)}(\Phi_1,\Phi_2,\Phi_3,\Phi_4) = -i\Big(\frac{\Omega_{f_4f_1}\Omega_{f_2f_3}}{s_{14}} + \frac{\Omega_{f_1f_2}\Omega_{f_3f_4}}{s_{12}}\Big)\delta^{(4)}(Q), \tag{5.29}$$

$$\mathcal{A}^{(1,0)}(\mathcal{V}_{1\dot{a}},\mathcal{V}_{2\dot{b}},\Phi_3,\Phi_4) = \frac{\Omega_{f_3f_4}}{s_{14}s_{12}}\delta^{(4)}(Q)[1_{\dot{a}}|\not{p}_4|2_{\dot{b}}]. \tag{5.30}$$

To verify that we have the correct superamplitudes, we can construct the four hyper superamplitude by applying the super-BCFW recursion relation on 3-point amplitudes with two hypers and one vector, with a vector as the propagator:

$$\begin{aligned}\mathcal{A}^{(1,0)}(\Phi_1,\Phi_2,\Phi_3,\Phi_4) &= -\frac{i}{s_{14}}\int d\eta_{pa}d\eta_p^a\mathcal{A}^{(1,0)}(\mathcal{V}_{p\dot{a}},\Phi_4,\Phi_{\hat{1}})\mathcal{A}^{(1,0)}(\mathcal{V}_{-p}^{\dot{a}},\Phi_{\hat{2}},\Phi_3)\\ &= \frac{i}{72}\frac{\Omega_{f_4f_1}\Omega_{f_2f_3}}{s_{14}}\int d\eta_{pa}d\eta_p^a\langle Q_LQ_LQ_L\rho\rangle\langle Q_RQ_RQ_R\rho\rangle(\tilde{u}_p.\tilde{u}_{-p}).\end{aligned}$$

where $Q_L = q_{\hat{1}} + q_4 + q_p$ and $Q_R = q_{\hat{2}} + q_3 + q_{-p}$. We can integrate the η_p's from either of the two $\langle\ldots\rangle$ and choose two of the three $Q_{R/L}$ which gives a factor of 3, and substituting $q_p = q_3 + q_{\hat{2}}$ ($q_p = q_{\hat{1}} + q_4$), we get (without the flavor/pole prefactor for convenience),

$$\begin{aligned}&= -\frac{1}{24}(\langle QQQ\rho\rangle\langle Q_L|p|\rho\rangle + \langle QQQ\rho\rangle\langle Q_R|p|\rho\rangle)(\tilde{u}_p.\tilde{u}_{-p})\\ &= -\frac{1}{24}\epsilon_{ABCD}Q^AQ^BQ^C\rho^D(\langle Q|^{A'}|p_{\dot{a}}]_{A'})[p^{\dot{a}}|\rho\rangle(\tilde{u}_p.\tilde{u}_{-p}).\end{aligned}$$

Here, we can notice that if $A' = A$ or B or C, the product of 4 Q vanishes. Thus, $A' = D$, i.e,

$$\begin{aligned}&= -\frac{1}{24}\epsilon_{ABCD}Q^AQ^BQ^CQ^D(\langle\rho|^{A'}|p_{\dot{a}}]_{A'})[p^{\dot{a}}|\rho\rangle(\tilde{u}_p.\tilde{u}_{-p})\\ &= -\frac{i}{4!}\langle Q^AQ^BQ^CQ^D\rangle(\tilde{w}_p.\tilde{w}_{-p})(\tilde{u}_p.\tilde{u}_{-p}).\end{aligned}$$

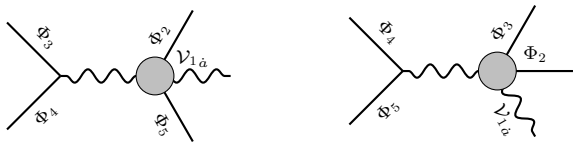
Finally using Eq.(B.3), we get

$$\mathcal{A}^{(1,0)}(\Phi_1,\Phi_2,\Phi_3,\Phi_4) = = -i\frac{\Omega_{f_4f_1}\Omega_{f_2f_3}}{s_{14}}\delta^{(4)}(Q).$$

Similar to the situation in section 4.2, there is another possible contribution from a flavor branch that we do not directly get from the super-BCFW recursion. Including that, the final expression of the super amplitude matches (5.29).

$$\mathcal{A}^{(1,0)}(\Phi_1,\Phi_2,\Phi_3,\Phi_4) = -i\Big(\frac{\Omega_{f_4f_1}\Omega_{f_2f_3}}{s_{14}} + \frac{\Omega_{f_1f_2}\Omega_{f_3f_4}}{s_{12}}\Big)\delta^{(4)}(Q).$$

6 One vector four hyper superamplitude



(a) (b)

Figure 6: Feynman diagrams for two possible flavor structures for the one vector four hyper superamplitude. The wavy line represents the vector multiplet and straight lines represent the matter multiplet. Figure 6a is the $\Omega_{f_3f_4}\Omega_{f_5f_2}$ branch and Figure 6b is the $\Omega_{f_2f_3}\Omega_{f_4f_5}$ branch.

For the one vector four hyper superamplitude with the color ordering (1,2,3,4,5), there are two possible flavor branches as shown in Fig. 6. As we saw in the calculation for four hyper superamplitude (Eq.(5.29)), using specific shifts for the BCFW recursion relation gives only one possible flavor branch from the diagrams. However, unlike the 4-point case, the other flavor branch is not just a cyclic permutation. Thus, to get the complete amplitude, one might have to compute another BCFW calculation for the other branch. Depending on the choice of the shifted legs, there is also the possibility of shifts that have a residue at $z \to \infty$. Instead of going through this tedious process, in this thesis we will devise an alternate bootstrap approach to compute amplitudes at higher multiplicities.

6.1 Bootstrap approach

In this section, we will construct the 5-point tree-level amplitude with one vector and four hypers as external states using a bootstrap approach. In this bootstrap approach, the amplitude is constructed in the following steps:

- Find linearly independent sets of numerators.
- Find the combination of numerators that satisfy the supersymmetric constraints $Q\mathcal{A} = 0$, $Q^\dagger\mathcal{A} = 0$.
- Construct a possible ansatz (with the proper dimensions) with Mandelstam variables in the denominator.
- Solve for free variables using amplitude relations.

We will use these steps to now construct the one vector four hyper superamplitude in the SQCD theory.

6.2 Possible numerators

To describe the helicity state of the vector particle, the numerator has to have an $SU(2)_R$ free index. WLOG, we can consider particle 1 to be the vector particle. Also, for five massless particles, there are only four independent momenta due to momentum conservation. Thus, any numerator with more than four momenta can be reduced to a simpler form with Mandelstam coefficients as follows

$$\langle q_1|\not{p}_2\not{p}_3\not{p}_4\not{p}_5|1_{\dot{a}}] = -\langle q_1|\not{p}_2\not{p}_3\not{p}_4\not{p}_2|1_{\dot{a}}] - \langle q_1|\not{p}_2\not{p}_3\not{p}_4\not{p}_3|1_{\dot{a}}]$$

$$= - s_{24}\langle q_1|\not{p}_2\not{p}_3|1_{\dot{a}}] - s_{23}\langle q_1|\not{p}_4\not{p}_2|1_{\dot{a}}] - s_{34}\langle q_1|\not{p}_2\not{p}_3|1_{\dot{a}}] \tag{6.1}$$

where we have used the anti-commutation relation $\{\not{p}_i, \not{p}_j\}^A_B = s_{ij}\delta^A_B$. Then, the possible Lorentz invariant numerators are

$$\langle q_i|1_{\dot{a}}]; \quad \langle q_i|\not{p}_j\not{p}_k|1_{\dot{a}}]. \tag{6.2}$$

Using momentum/supermomentum conservation, we can count ten possible numerators. However, numerically checking in Mathematica, we finally have six linearly independent numerators. Including the $\delta^{(4)}(Q)$ for supermomentum conservation, the numerators are

$$\begin{aligned}&\delta^{(4)}(Q)\langle q_2|1_{\dot{a}}], \quad \delta^{(4)}(Q)\langle q_3|1_{\dot{a}}], \quad \delta^{(4)}(Q)\langle q_4|1_{\dot{a}}];\\ &\delta^{(4)}(Q)\langle q_1|\not{p}_2\not{p}_3|1_{\dot{a}}], \quad \delta^{(4)}(Q)\langle q_1|\not{p}_2\not{p}_4|1_{\dot{a}}], \quad \delta^{(4)}(Q)\langle q_2|\not{p}_3\not{p}_4|1_{\dot{a}}].\end{aligned} \tag{6.3}$$

6.3 Constraints on the amplitude

Supersymmetric Ward identities

In any supersymmetric theory, the superamplitude ($\mathcal{A}$) vanishes under the action of the supercharges Q [33]. They also vanish under the action of the hermitian conjugate of the supercharge $Q^\dagger$. For the Q defined as $\sum_i \lambda_{ia}\eta^a_i$, the $Q^\dagger$ is defined as

$$\sum_i \lambda_{ia}\frac{\partial}{\partial\eta_{ia}}. \tag{6.4}$$

From our linearly independent set, we will use these identities to solve for functions that act as a basis for our amplitudes.

The first constraint $Q\mathcal{A} = 0$ is automatically satisfied by the supermomentum delta function $\delta^{(4)}(Q)$. However, the second constraint $Q^\dagger\mathcal{A} = 0$ is non-trivial. While it is true that $Q^\dagger\delta^{(4)}(Q) = 0$ due to the Leibniz product rule, $Q^\dagger$ also acts on whatever follows the delta function. Thus, one can solve this constraint to get an idea about the possible functions that can be part of the amplitude. Solving this numerically, we got the basis functions

$$f[i,j] = -\frac{\langle q_i|1_{\dot{a}}]}{s_{1i}} + \frac{\langle q_j|1_{\dot{a}}]}{s_{1j}} + \frac{\langle q_1|\not{p}_i\not{p}_j|1_{\dot{a}}]}{s_{1i}s_{1j}} \tag{6.5}$$

where i, j are particle indices and can be particles $2, 3, 4, 5$.

Before we construct an ansatz and solve for the amplitudes, we can study the properties of the basis function f's. It is straightforward to see that it is antisymmetric under the exchange of i, j. The other properties that are not obvious were deduced numerically, and are listed below.

$$f[i,j] = -f[j,i,] \tag{6.6a}$$
$$f[i,j] + f[j,k] = f[i,k], \tag{6.6b}$$
$$s_{1j}f[i,j] + s_{1k}f[i,k] + s_{1l}f[i,l] = 0. \tag{6.6c}$$

These properties give us a total of four independent constraints. Therefore, there are a total of two independent f's which can now be used as a part of the ansatz for the superamplitude.

Amplitude relations

All color-ordered tree-level amplitudes (as well as superamplitudes) are known to satisfy some relations that reduce the number of independent color-ordered amplitudes [34, 35]. The most basic relations are the cyclic relation and the mirror relation, which are given due to the properties of the trace prefactor in Eq.(2.9). They are given by

$$A(1,2,\ldots,n-1,n) = A(2,3,\ldots,n,1), \tag{6.7a}$$
$$A(1,2,\ldots,n-1,n) = (-1)^n A(n,n-1,\ldots,2,1). \tag{6.7b}$$

One non-trivial relation is the $U(1)$ decoupling relation (also known as the sub-cyclic relation):

$$\sum_{\sigma\in\text{cyclic}} \mathcal{A}_n(1,\sigma(2,3,\ldots,n)) = 0 \tag{6.8}$$

where one leg is fixed, and we cycle over the other legs. This relation is derived by replacing $T^{a_1} \to 1$ (i.e. the $U(1)$ generataor) in Eq.(2.9). Since there is no interaction between the $U(1)$ particle and the other fields, this sum has to vanish. This relation is also a special case of the Kleiss-Kuijf (KK) relation [34], whose constraints reduce the number of independent $n-$point color-ordered diagrams to $(n-2)!$, known as the KK basis.

At the 5-point level the $U(1)$ decoupling relation is given by

$$\mathcal{A}_5(1,2,3,4,5) + \mathcal{A}_5(1,3,4,5,2) + \mathcal{A}_5(1,4,5,2,3) + \mathcal{A}_5(1,5,2,3,4) = 0 \tag{6.9}$$

Another set of relations was discovered through color kinematics duality by Z. Bern, J. J. M. Carrasco and H. Johansson (BCJ) in [35]. This duality states that a gauge theory amplitude can be written in the form,

$$\mathcal{A}_n = \sum_i \frac{c_i n_i}{d_i} \tag{6.10}$$

where c_i are the color factors, n_i are the kinematic numerators, and d_i are the poles. Moreover, they conjecture that the kinematic numerators obey the same relations as their color prefactors, particularly the Jacobi-like relations known as the BCJ relations

$$c_i - c_j = c_k \implies n_i - n_j = n_k. \tag{6.11}$$

The BCJ relations linear in Mandelstam variables are given as

$$\sum_{i=2}^{n-1}(\sum_{j=2}^{i} s_{nj})\mathcal{A}_n(1,2,\ldots,i,n,i+1,\ldots,n-1) = 0. \tag{6.12}$$

Using such relations, one can reduce the number of independent $n-$point color ordered diagrams to $(n-3)!$, known as the BCJ basis. At the 5-point level, with the vector particle (1) being the permuted leg the relation is

$$s_{15}\mathcal{A}_5(1,2,3,4,5) + (s_{15}+s_{12})\mathcal{A}_5(1,3,4,5,2) + (s_{12}+s_{13}+s_{15})\mathcal{A}_5(1,4,5,2,3) = 0. \tag{6.13}$$

6.4 Ansatz

The 5-point amplitude ansatz needs to be of mass dimension -1. The function f's have mass-dimension 1, but, before using them in the ansatz, we make the poles that appear in the f's local by multiplying $s_{1i}s_{1j}$. This leaves them with mass-dimensions 5. So to have mass-dimension -1, we need to have three Mandelstam variables (each mass-dimension 2) as poles in the denominator.

For the color-ordered amplitude, the only planar diagrams are allowed. This means that only the consecutive poles $s_{12}, s_{23}, s_{34}, s_{45}, s_{51}$ are part of the Feynman diagrams. For 5-point Mandelstam variables, these form a linearly independent basis. For a coefficient with three denominators, we can have 10 possible combinations.

There are also two possible flavors for the planar diagrams, $\Omega_{23}\Omega_{45}$ and $\Omega_{34}\Omega_{52}$. Along with two basis functions, we have an ansatz with $2 \times 2 \times \binom{5}{3} = 40$ coefficients that need to be fixed. Before fixing the coefficients using the amplitude relations, we need to choose the two linearly independent functions from the 6 basis functions. It turns out some choices of functions lead to unphysical poles in the denominator. To illustrate, look at two possible choices below.

$$
\begin{aligned}
&\textbf{Choice A: } f[2,3] \text{ and } f[4,5]\\
&\qquad f[3,4] = \frac{s_{15}f[4,5] - s_{12}f[2,3]}{s_{12}+s_{13}},\\
&\qquad f[2,4] = \frac{s_{13}f[2,3] + s_{15}f[4,5]}{s_{12}+s_{13}},\\
&\qquad f[3,5] = \frac{s_{14}f[4,5] + s_{12}f[2,3]}{s_{14}+s_{15}},\\
&\qquad f[2,5] = \frac{s_{14}f[4,5] - s_{13}f[2,3]}{s_{14}+s_{15}}.\\
&\textbf{Choice B: } f[2,3] \text{ and } f[2,4]\\
&\qquad f[3,4] = -f[2,3] + f[2,4],\\
&\qquad f[2,5] = \frac{s_{13}f[2,3] + s_{14}f[4,5]}{s_{15}},\\
&\qquad f[3,5] = \frac{s_{13}f[2,3] + (s_{45}-s_{23})f[4,5]}{s_{15}},\\
&\qquad f[4,5] = \frac{s_{13}f[2,3] - s_{14}f[2,3]}{s_{15}}.
\end{aligned}
$$

In choice A, we see that when we rewrite the other functions in terms of $f[2,3], f[4,5]$, there are unphysical poles in the denominator. However, in choice A, only s_{15} appears in the denominator, which is a physical pole. Thus, we choose basis choice B in the ansatz.

The final ansatz for the one vector four hyper superamplitude is

$$
\begin{aligned}
\mathcal{A}_5 =&\Omega_{23}\Omega_{45}s_{12}s_{13}f[2,3]\sum\frac{A_i}{s_{i_1i_2}s_{j_1j_2}s_{k_1k_2}} + \Omega_{52}\Omega_{34}s_{12}s_{13}f[2,3]\sum\frac{B_i}{s_{i_1i_2}s_{j_1j_2}s_{k_1k_2}}\\
&+\Omega_{23}\Omega_{45}s_{12}s_{14}f[2,4]\sum\frac{C_i}{s_{i_1i_2}s_{j_1j_2}s_{k_1k_2}} + \Omega_{52}\Omega_{34}s_{12}s_{14}f[2,4]\sum\frac{D_i}{s_{i_1i_2}s_{j_1j_2}s_{k_1k_2}}. \quad (6.14)
\end{aligned}
$$

To fix the coefficients, we will use the amplitude relations Eq.(6.9) and Eq.(6.13). Solving for the coefficients with all possible color-ordered equations of these two types, we get the final superamplitude as

$$\mathcal{A}_5(\mathcal{V}_{1\dot{a}},\Phi_2,\Phi_3,\Phi_4,\Phi_5)=i\Omega_{f_2f_3}\Omega_{f_4f_5}\delta^{(4)}(Q)(\frac{s_{13}f[2,3]}{s_{23}s_{45}}-\frac{f[2,5]}{s_{23}})-\Omega_{f_5f_2}\Omega_{f_3f_4}\delta^{(4)}(Q)(\frac{f[2,5]}{s_{34}}). \tag{6.15}$$

Verification

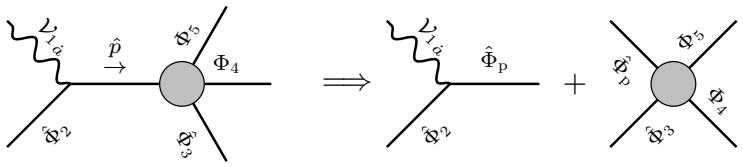


Figure 7: Factorization of the one vector four hyper superamplitude into one vector two hyper superamplitude and four hyper superamplitude under the limit $s_{1\hat{2}}=\hat{p}^2\to 0$.

To fix the overall normalization and make sure that we have the right amplitude, we can take the limit where the propagator vanishes to make sure that the amplitude factorizes correctly [22, 23]. Numerically, it is easy to do this verification by implementing the BCFW shifts (Eq. (4.11)) to make a propagator vanish.

By shifting the 2,3 legs, we can have $s_{1\hat{2}}=0$. Under this limit, it was verified that the amplitude factorizes into the correct 3-point and 4-point amplitudes (see Fig. 7).

$$s_{1\hat{2}}\mathcal{A}_5(V_{1\dot{a}},\hat{\Phi}_2,\hat{\Phi}_3,\Phi_4,\Phi_5)=i\mathcal{A}_3(V_{1\dot{a}},\hat{\Phi}_2,\Phi_p)\times\mathcal{A}_4(\Phi_p,\hat{\Phi}_3,\Phi_4,\Phi_5). \tag{6.16}$$

The same verification was also done for other propagators to make sure they factorize correctly.

7 Supergravity

Similar to gauge theories, one can construct supergravity theories by including the background curvature/ Ricci scalar in the Lagrangian that gives rise to gravity-mediating particles such as gravitons and their fermionic partners, gravitinos. Unlike a generic gauge theory that has only 3- or 4-point vertices, the gravity interactions in the supergravity Lagrangian give rise to an infinite number of vertices, which lead to more complex interactions. While it is possible to simplify the Feynman rules using specific gauge choices, calculations beyond leading order are still very complicated.

In this section, we will introduce the double copy framework, which lets us glean gravity amplitudes from two copies of gauge theory amplitudes. The double copy structure was discovered by Kawai, Lewellen, and Tye (KLT) in 1985, when they showed that closed-string tree amplitudes can be written as sums of products of open string tree amplitudes [18]. At the field theory limit, it gives a relation between gravity amplitudes as sums of products of pairs of gauge amplitudes. These KLT relations take the form,

$$\mathcal{M}_n=-i\sum_{\sigma,\rho\in S_{n-3}}\mathcal{A}_n(1,\sigma,n-1,n)S[\sigma|\rho]\tilde{A}_n(1,\rho,n,n-1) \tag{7.1}$$

where $S[\sigma|\rho]$ is the KLT kernel, which is given as a function of the Mandelstam variables.

In 2010, the duality between color and kinematics was used to derive new relations for gravity amplitudes from gauge theory amplitudes [36] known as the BCJ double copy relations. This duality states that for a gauge theory amplitude of the form Eq. (6.10) that satisfies Eq. (6.11), one can construct gravity amplitudes by a simple replacement of the color factors with the kinematics numerator:

$$\mathcal{M}_n = \sum_i \frac{n_i \tilde{n}_i}{d_i}. \tag{7.2}$$

Similar to these relations at the amplitude level, the double copy can also be used to construct the field content in the corresponding gravity theory from component gauge theories. For example, the polarization vectors of the fields in a pure-gravity theory can be mapped to those of the component pure-YM theory as

$$\begin{aligned} &\text{graviton: } \epsilon_\mu^{(i}\epsilon_\nu^{j)}, \\ &\text{B-field: } \epsilon_\mu^{[i}\epsilon_\nu^{j]}, \\ &\text{dilaton: } \epsilon_\mu^{i}\epsilon_\nu^{j}\delta_{ij}, \end{aligned}$$

where i, j are the little group indices.

For the case of this thesis, we have a QCD theory, where in addition to a vector field, we also have fermionic fields such as gauginos. Under the double copy construction, these map to fields such as

$$\begin{aligned} &\text{gravitino: vector boson} \otimes \text{gaugino}, \\ &\text{graviphoton: gaugino} \otimes \text{gaugino}, \\ &\text{matter fields: gaugino} \otimes \text{scalar, scalar} \otimes \text{scalar}. \end{aligned}$$

This double copy structure also holds for the superfields in the gravity theory as a double copy of the gauge theory superfields. Specifically, for the $\mathcal{N} = (1,0)$ (or equivalently $\mathcal{N} = (0,1)$) SQCD theory, one can construct the half-maximal supergravity theories as shown in Table 1.

	$\mathcal{N} = (1,0)$ SQCD	$\mathcal{N} = (0,1) SQCD$
$\mathcal{N} = (1,0)$ SQCD	$\mathcal{N} = (2,0)$ $\mathcal{H}^{(2,0)} \oplus \mathcal{T}^{(2,0)} = \mathcal{V}^{(1,0)} \otimes \mathcal{V}^{(1,0)}$ $\mathcal{T}^{(2,0)} = \Phi^{(1,0)} \otimes \Phi^{(1,0)}$	$\mathcal{N} = (1,1)$ $\mathcal{H}^{(1,1)} = \mathcal{V}^{(1,0)} \otimes \mathcal{V}^{(0,1)}$ $\mathcal{V}^{(1,1)} = \Phi^{(1,0)} \otimes \bar{\Phi}^{(0,1)}$
$\mathcal{N} = (0,1)$ SQCD	$\mathcal{N} = (1,1)$ $\mathcal{H}^{(1,1)} = \mathcal{V}^{(0,1)} \otimes \mathcal{V}^{(1,0)}$ $\mathcal{V}^{(1,1)} = \Phi^{(0,1)} \otimes \bar{\Phi}^{(1,0)}$	$\mathcal{N} = (0,2)$ $\mathcal{H}^{(0,2)} \oplus \mathcal{T}^{(0,2)} = \mathcal{V}^{(0,1)} \otimes \mathcal{V}^{(0,1)}$ $\mathcal{T}^{(0,2)} = \Phi^{(0,1)} \otimes \Phi^{(0,1)}$

Table 1: Possible supergravity theories and their corresponding supermultiplets as double copy of $\mathcal{N} = (1,0)$ and $\mathcal{N} = (0,1)$ SQCD in 6D. The theories in the diagonal give the same theory ($\mathcal{N} = (1,1)$) due to the symmetric structure of the double copy. The off-diagonal theories are equivalent with the exception of being chiral ($\mathcal{N} = (2,0)$) and anti-chiral $\mathcal{N} = (0,2)$.

7.1 $\mathcal{N}$=(2,0) Chiral supergravity

Taking two copies of $\mathcal{N} = (1,0)$ SQCD theory (or equivalently $\mathcal{N} = (0,2)$ from two copies of $\mathcal{N} = (0,1)$), we get the half maximal $\mathcal{N} = (2,0)$ chiral supergravity theory. This theory has 16 supercharges, $q^A_{a\alpha}$, where A is the Lorentz spinor index, a is the little group index, and α is the R-symmetry index.

This theory comprises tensor multiplets and graviton multiplets as shown below (with the R-symmetry indices explicitly expanded):

$$\mathcal{V}^{(1,0)}_{\dot{a}} \otimes \mathcal{V}^{(1,0)}_{\dot{b}} = \mathcal{H}^{(2,0)}_{(\dot{a}\dot{b})} \oplus \epsilon_{\dot{a}\dot{b}} \mathcal{T}^{(2,0)}, \tag{7.3a}$$

$$\mathcal{T}^{(2,0)} = \Phi^{(1,0)} \otimes \Phi^{(1,0)} \tag{7.3b}$$

$$\begin{aligned}\mathcal{H}^{(2,0)}_{\dot{a}\dot{b}} =& B^1_{\dot{a}\dot{b}} + \eta_{1a}\psi^{1a}_{\dot{a}\dot{b}} + \eta_{2a}\psi^{2a}_{\dot{a}\dot{b}} + (\eta_1)^2 B^2_{\dot{a}\dot{b}} + (\eta_2)^2 B^3_{\dot{a}\dot{b}} + (\eta_1.\eta_2) B^4_{\dot{a}\dot{b}} \\ &+ \eta_{1a}\eta_{2b} G^{ab}_{\dot{a}\dot{b}} + (\eta_1)^2 \eta_{2a}\psi^{3a}_{\dot{a}\dot{b}} + (\eta_2)^2 \eta_{1a}\psi^{4a}_{\dot{a}\dot{b}} + (\eta^1)^2(\eta^2)^2 B^5_{\dot{a}\dot{b}}\end{aligned} \tag{7.3c}$$

$$\begin{aligned}\mathcal{T}^{(2,0)} =& \phi_1 + \chi^a_1 \eta^1_a + \chi^a_2 \eta^2_a + (\eta_1)^2 \phi_2 + (\eta_2)^2 \phi_3 + (\eta_1.\eta_2)\phi_4 \\ &+ \eta^{1a}\eta^{2b} B_{(ab)} + (\eta^1)^2 \eta^2_a \tilde{\chi}^a_2 + (\eta^2)^2 \eta^1_a \tilde{\chi}^a_1 + (\eta^1)^2(\eta^2)^2 \phi_5\end{aligned} \tag{7.3d}$$

The graviton multiplet has 24 bosonic and 24 fermionic DoF. The bosonic fields consists of a graviton field (9 DoF) and five self-dual B-fields (15 DoF), while the fermionic fields are four gravitino fields with 6 DoF each.

The matter multiplet in this theory is the tensor multiplet with eight bosonic (1 self-dual B-field + 5 scalars) and eight fermionic DoF (2 SMW fermions).

Similar to the $\mathcal{N} = (1,0)$ SQCD theory, there can be N_t copies of the tensor multiplet in the $\mathcal{N} = (2,0)$ supergravity. This gives rise to a flavor symmetry of $SO(N_t)$ in the theory. The invariant form in the $SO(N_t)$ flavor group is given by the symmetric delta function δ_{ij}.

7.2 $\mathcal{N}$=(1,1) Non-chiral supergravity

The non-chiral $\mathcal{N} = (1,1)$ supergravity can be realized as the double copy of $\mathcal{N} = (1,0) \otimes \mathcal{N} = (0,1)$ SQCD theories. This theory also has 16 supercharges, q^A_a, where A is the Lorentz spinor index, a is the $SU(2)_L$ little group index and $\tilde{q}_{A\dot{a}}$, where $\dot{a}$ is the $SU(2)_L$ little group index.

The matter in this theory is encapsulated as a vector multiplet alongside a graviton multiplet as shown below:

$$\mathcal{V}^{(0,1)}_{a} \otimes \mathcal{V}^{(1,0)}_{\dot{a}} = \mathcal{H}^{(1,1)}_{a\dot{a}}, \tag{7.4a}$$

$$\mathcal{V}^{(1,1)} = \Phi^{(1,0)} \otimes \bar{\Phi}^{(0,1)} \tag{7.4b}$$

$$\begin{aligned}\mathcal{H}^{(1,1)}_{b\dot{b}} =& g^1_{b\dot{b}} + \eta^a(\psi^1_{ab\dot{b}} + \frac{\epsilon_{ab}\tilde{\chi}^1_{\dot{b}}}{\sqrt{2}}) + \tilde{\eta}^{\dot{a}}(\tilde{\psi}^1_{\dot{a}b\dot{b}} + \frac{\epsilon_{\dot{a}\dot{b}}\chi^1_b}{\sqrt{2}}) + \eta^a\tilde{\eta}^{\dot{a}}(G_{a\dot{a}b\dot{b}} + \frac{\epsilon_{ab}B_{\dot{a}\dot{b}}}{\sqrt{2}} + \frac{\epsilon_{\dot{a}\dot{b}}B_{ab}}{\sqrt{2}} + \frac{\epsilon_{ab}\epsilon_{\dot{a}\dot{b}}\phi}{2}) \\ &+ (\eta)^2 g^2_{b\dot{b}} + (\tilde{\eta})^2 g^3_{b\dot{b}} + (\tilde{\eta})^2\eta^a(\psi^2_{ab\dot{b}} + \frac{\epsilon_{ab}\tilde{\chi}^2_{\dot{b}}}{\sqrt{2}}) + (\eta)^2\tilde{\eta}^{\dot{a}}(\tilde{\psi}^2_{\dot{a}b\dot{b}} + \frac{\epsilon_{\dot{a}\dot{b}}\chi^2_b}{\sqrt{2}}) + (\eta)^2(\tilde{\eta})^2 g^4_{b\dot{b}}\end{aligned} \tag{7.4c}$$

$$\mathcal{V}^{(1,1)} = \phi^1 + \eta_a\chi^a + \tilde{\eta}^{\dot{a}}\tilde{\chi}_{\dot{a}} + \eta_a\tilde{\eta}^{\dot{a}} g^a{}_{\dot{a}} + (\eta)^2\phi^2 + (\tilde{\eta})^2\phi^3 + (\eta)^2\tilde{\eta}^{\dot{a}}\bar{\tilde{\chi}}_{\dot{a}} + (\tilde{\eta})^2\eta_a\bar{\chi}^a + (\eta)^2(\tilde{\eta})^2\phi^4 \tag{7.4d}$$

The graviton multiplet has 32 bosonic and 32 fermionic DoF. The bosonic fields consist of a graviton field (9 DoF), two B-fields (6 Dof), four gravi-photons (16 DoF), and one dilaton (1 DoF) while the fermionic fields are four gravitino fields (24 DoF) with two Weyl fields (8 DoF).

The vector multiplet has eight bosonic DoF (1 vector + 3 scalars) and eight fermionic DoF (2 Weyl fermions).

For N_v copies of the vector multiplet, this theory has a flavor symmetry of $SO(N_v)$.

7.3 Amplitudes

Now that we have seen the maps of the supergravity superfields to the SQCD superfields, we can construct the required superamplitudes in the supergravity theory using the double copy structure. While the previous description of the double copy holds when all particles are in the adjoint representation, there are some nuances to consider with flavored matter. Consider the KLT relation Eq.(7.1), here σ, ρ consider all possible permutations of the particles (2,. . .,n-2). However, when particles are in the fundamental representation, some permutations will be forbidden by flavor. Thus, to get the matter amplitudes in the supergravity theory, we will follow the description given in [37].

3-point

The KLT kernel at the 3-point level is trivial as $\sigma, \rho = \emptyset$. Therefore, the matter amplitude is just given as

$$\mathcal{M}_3 = i\mathcal{A}(1, \underline{2}, \overline{3})\tilde{\mathcal{A}}(1, \underline{2}, \overline{3}). \tag{7.5}$$

Here we introduce the use of $\{\underline{2}, \overline{3}\}$ to show matter pairs that are of the same flavor.

The one graviton two matter amplitude in the $\mathcal{N} = (2, 0)$ supergravity is given as

$$\begin{aligned}\mathcal{M}^{(2,0)}(\mathcal{H}_{1\dot{a}\dot{b}}, \mathcal{T}_2, \mathcal{T}_3) =& - i\mathcal{A}(\mathcal{V}_{1\dot{a}}, \Phi_2, \Phi_3) \odot \mathcal{A}(\mathcal{V}_{1\dot{b}}, \Phi_2, \Phi_3) \\ =& i\frac{\delta_{f_2 f_3}}{(3!)^2} \prod_{\alpha=\{1,2\}} \langle Q_\alpha Q_\alpha Q_\alpha \rho\rangle \tilde{u}_{(1\dot{a}} \tilde{u}_{1\dot{b})}\end{aligned} \tag{7.6}$$

where, $\odot$ implies symmetric tensor product, α is the R-symmetry index and the (.) brackets on little group index implies symmetrization.

Similarly, in the $\mathcal{N} = (1, 1)$ supergravity,

$$\begin{aligned}\mathcal{M}^{(1,1)}(\mathcal{H}_{1a\dot{a}}, \mathcal{V}_2, \mathcal{V}_3) =& - i\mathcal{A}(\mathcal{V}_{1a}, \Phi_2, \Phi_3) \odot \mathcal{A}(\mathcal{V}_{1\dot{a}}, \bar{\Phi}_2, \bar{\Phi}_3) \\ =& i\frac{\delta_{f_2 f_3}}{(3!)^2} \langle QQQ\rho\rangle [\tilde{Q}\tilde{Q}\tilde{Q}\tilde{\rho}] u_{1a} \tilde{u}_{1\dot{a}}.\end{aligned} \tag{7.7}$$

4 point

At the 4-point level, $\sigma, \rho = 2$. Thus, once we fix the flavor choice $\{\underline{1}, \overline{2}\}, \{\underline{3}, \overline{4}\}$, the KLT kernel remains the same as the adjoint one. i.e.,

$$\mathcal{M}_4 = -i\mathcal{A}(\underline{1}, \overline{2}, \underline{3}, \overline{4})\tilde{\mathcal{A}}(\underline{1}, \overline{2}, \underline{4}, \overline{3})s_{12}. \tag{7.8}$$

However, for the total amplitude, we have to consider all possible flavor choices, which would include the permutations of the above relation.

For the 4-point amplitude in the supergravity theories, we can have either one pair or two pairs of matter multiplets as external states. The corresponding amplitudes in the $\mathcal{N}=(2,0)$ are given as

$$\begin{aligned}\mathcal{M}^{(2,0)}(\mathcal{H}_{1\dot{a_1}\dot{a_2}},\mathcal{H}_{2\dot{b_1}\dot{b_2}},\mathcal{T}_3,\mathcal{T}_4) =& -i\mathcal{A}(\mathcal{V}_{1\dot{a_1}},\mathcal{V}_{2\dot{b_1}},\Phi_3,\Phi_4)\odot\tilde{\mathcal{A}}(\mathcal{V}_{1\dot{a_2}},\mathcal{V}_{2\dot{b_2}},\Phi_4,\Phi_3)s_{12}\\ =& -i\delta_{f_3f_4}\delta^{(4)}(Q_1)\delta^{(4)}(Q_2)\frac{[1_{(\dot{a_1}}|\not{p}_3|2_{(\dot{b_1}}][1_{\dot{a_2})}|\not{p}_3|2_{\dot{b_2})}]}{s_{12}s_{13}s_{14}},\end{aligned} \tag{7.9}$$

$$\begin{aligned}\mathcal{M}^{(2,0)}(\mathcal{T}_1,\mathcal{T}_2,\mathcal{T}_3,\mathcal{T}_4) =& -i\mathcal{A}(\Phi_1,\Phi_2,\Phi_3,\Phi_4)\odot\tilde{\mathcal{A}}(\Phi_1,\Phi_2,\Phi_4,\Phi_3)s_{12}\\ =& i\delta_{f_1f_2}\delta_{f_3f_4}\frac{\delta^{(4)}(Q_1)\delta^{(4)}(Q_2)}{s_{12}}.\end{aligned} \tag{7.10}$$

In the $\mathcal{N}=(1,1)$ supergravity,

$$\begin{aligned}\mathcal{M}^{(1,1)}(\mathcal{H}_{1a\dot{a}},\mathcal{H}_{2b\dot{b}},\mathcal{V}_3,\mathcal{V}_4) =& -i\mathcal{A}(\mathcal{V}_{1a},\mathcal{V}_{2b},\Phi_3,\Phi_4)\odot\tilde{\mathcal{A}}(\mathcal{V}_{1\dot{a}},\mathcal{V}_{2\dot{b}},\bar{\Phi}_4,\bar{\Phi}_3)s_{12}\\ =& -i\delta_{f_3f_4}\delta^{(4)}(Q)\delta^{(4)}(\tilde{Q})\frac{\langle 1_a|\not{p}_3|2_b\rangle[1_{\dot{a}}|\not{p}_3|2_{\dot{b}}]}{s_{12}s_{13}s_{14}},\end{aligned} \tag{7.11}$$

$$\begin{aligned}\mathcal{M}^{(1,1)}(\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3,\mathcal{V}_4) =& -i\mathcal{A}(\Phi_1,\Phi_2,\Phi_3,\Phi_4)\odot\tilde{\mathcal{A}}(\Phi_1,\Phi_2,\Phi_4,\Phi_3)s_{12}\\ =& i\delta_{f_1f_2}\delta_{f_3f_4}\frac{\delta^{(4)}(Q)\delta^{(4)}(\tilde{Q})}{s_{12}}.\end{aligned} \tag{7.12}$$

5-point

The 5-point amplitude is where the relations get nontrivial. Following [37], we choose the matter pairs to be $\{\underline{1},\overline{2}\},\{\underline{3},\overline{4}\}$. The formula for two matter pairs at n point is given as

$$\mathcal{M}_n=-i\sum_{\alpha,\beta\in S_{n-3}(\{\overline{4},\ldots,n\})}\mathcal{A}(\underline{1},\overline{2},\underline{3},\alpha)\tilde{\mathcal{A}}(\overline{2},\underline{1},\underline{3},\beta)\prod_{i=1}^{n-3}\Big(\sum_{j\in A\cap B}s_{\alpha_i j}\Big) \tag{7.13}$$

where $A=\{3,\alpha_1,\ldots,\alpha_{i-1}\}$,$B=\{3,\beta_1,\ldots,\beta_{b_i-1}\}$ and $\alpha_i=\beta_{b_i}$. At 5-point, this expands to

$$\begin{aligned}\mathcal{M}_5 =& -i\sum_{\alpha,\beta\in S_2(\{\overline{4},5\})}\mathcal{A}(\underline{1},\overline{2},\underline{3},\alpha)\tilde{\mathcal{A}}(\overline{2},\underline{1},\underline{3},\beta)\prod_{i=1}^{2}\Big(\sum_{j\in A\cap B}s_{\alpha_i j}\Big)\\ =& i\begin{pmatrix}\mathcal{A}(\underline{1},\overline{2},\underline{3},\overline{4},5) & \mathcal{A}(\underline{1},\overline{2},\underline{3},5,\overline{4})\end{pmatrix}\begin{pmatrix}s_{34}(s_{35}+s_{45}) & s_{34}s_{35}\\ s_{35}s_{34} & s_{35}(s_{34}+s_{45})\end{pmatrix}\begin{pmatrix}\tilde{\mathcal{A}}(\overline{2},\underline{1},\underline{3},\overline{4},5)\\ \tilde{\mathcal{A}}(\overline{2},\underline{1},\underline{3},5,\overline{4})\end{pmatrix}.\end{aligned} \tag{7.14}$$

Substituting it for the $\mathcal{N}=(2,0)$, we get

$$\begin{aligned}&\mathcal{M}^{(2,0)}(\mathcal{T}_1,\mathcal{T}_2,\mathcal{T}_3,\mathcal{T}_4,\mathcal{H}_{5\dot{a}\dot{b}})\\ &=i\mathcal{A}(\Phi_1,\Phi_2,\Phi_3,\Phi_4,V_{5\dot{a}})\odot\tilde{\mathcal{A}}(\Phi_2,\Phi_1,\Phi_3,\Phi_4,V_{5\dot{b}})s_{34}(s_{35}+s_{45})\\ &\quad+i\mathcal{A}(\Phi_1,\Phi_2,\Phi_3,\Phi_4,V_{5\dot{a}})\odot\tilde{\mathcal{A}}(\Phi_2,\Phi_1,\Phi_3,V_{5\dot{b}},\Phi_4)s_{34}s_{35}\\ &\quad+i\mathcal{A}(\Phi_1,\Phi_2,\Phi_3,V_{5\dot{a}},\Phi_4)\odot\tilde{\mathcal{A}}(\Phi_2,\Phi_1,\Phi_3,\Phi_4,V_{5\dot{b}})s_{35}s_{34}\end{aligned}$$

$$
\begin{aligned}
&+ i\mathcal{A}(\Phi_1,\Phi_2,\Phi_3,V_{5\dot{a}},\Phi_4)\odot\tilde{\mathcal{A}}(\Phi_2,\Phi_1,\Phi_3,V_{5\dot{b}},\Phi_4)s_{35}(s_{34}+s_{45})\\
=&i\delta_{f_1f_2}\delta_{f_3f_4}\delta^{(4)}(Q_1)\delta^{(4)}(Q_2)\\
&\Big(\Big(\frac{f_{(1\dot{a}}[1,4]}{s_{12}}-\frac{f_{(1\dot{a}}[1,2]s_{25}}{s_{12}s_{34}}\Big)\Big(\frac{f_{2\dot{b})}[2,4]}{s_{12}}-\frac{f_{2\dot{b})}[2,1]s_{15}}{s_{12}s_{34}}\Big)s_{34}(s_{35}+s_{45})\\
&-\Big(\frac{f_{(1\dot{a}}[1,4]}{s_{12}}-\frac{f_{(1\dot{a}}[1,2]s_{25}}{s_{12}s_{34}}\Big)\frac{f_{2\dot{b})}[4,3]}{s_{12}}s_{34}s_{35}\\
&-\frac{f_{(1\dot{a}}[4,3]}{s_{12}}\Big(\frac{f_{2\dot{b})}[2,4]}{s_{12}}-\frac{f_{2\dot{b})}[2,1]s_{15}}{s_{12}s_{34}}\Big)s_{35}s_{34}+\frac{f_{(1\dot{a}}[4,3]}{s_{12}}\frac{f_{2\dot{b})}[4,3]}{s_{12}}s_{35}(s_{34}+s_{45})\Big)\\
=&i\delta_{f_1f_2}\delta_{f_3f_4}\delta^{(4)}(Q_1)\delta^{(4)}(Q_2)\\
&\Big(\Big(\frac{f_{(1\dot{a}}[1,4]f_{2\dot{b})}[1,2]}{s_{12}}-\frac{f_{(1\dot{a}}[1,2]f_{2\dot{b})}[1,2]s_{25}}{s_{12}s_{34}}\Big)s_{15}+\frac{f_{(1\dot{a}}[4,3]\tilde{f}_{2\dot{b})}[2,3]}{s_{12}}s_{35}\Big)
\end{aligned}
\tag{7.15}
$$

where

$$
f_{\alpha\dot{a}}[i,j]=-\frac{\langle q_{\alpha i}|5_{\dot{a}}]}{s_{5i}}+\frac{\langle q_{\alpha j}|5_{\dot{a}}]}{s_{5j}}+\frac{\langle q_{\alpha 5}|\not{p}_i\not{p}_j|5_{\dot{a}}]}{s_{5i}s_{5j}},
\tag{7.16}
$$

$$
\tag{7.17}
$$

and α is the R-symmetry index.

Similarly for $\mathcal{N}=(1,1)$ supergravity, we get the final amplitude as

$$
\begin{aligned}
&\mathcal{M}^{(1,1)}(\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3,\mathcal{V}_4,\mathcal{H}_{5a\dot{a}})\\
&\quad=i\delta_{f_1f_2}\delta_{f_3f_4}\delta^{(4)}(Q)\delta^{(4)}(\tilde{Q})\\
&\qquad\Big(\Big(\frac{f_{\dot{a}}[1,4]}{s_{12}}-\frac{f_{\dot{a}}[1,2]s_{25}}{s_{12}s_{34}}\Big)\tilde{f}_a[1,2]s_{15}+\frac{f_{\dot{a}}[4,3]}{s_{12}}\tilde{f}_a[2,3]s_{35}\Big)
\end{aligned}
\tag{7.18}
$$

where

$$
\begin{aligned}
f_{\dot{a}}[i,j]&=-\frac{\langle q_i|5_{\dot{a}}]}{s_{5i}}+\frac{\langle q_j|5_{\dot{a}}]}{s_{5j}}+\frac{\langle q_5|\not{p}_i\not{p}_j|5_{\dot{a}}]}{s_{5i}s_{5j}},\\
\tilde{f}_a[i,j]&=-\frac{\langle 5_a|\tilde{q}_i]}{s_{5i}}+\frac{\langle 5_a|\tilde{q}_j]}{s_{5j}}+\frac{\langle 5_a|\not{p}_i\not{p}_j|\tilde{q}_5]}{s_{5i}s_{5j}}.
\end{aligned}
\tag{7.19}
$$

8 Dimensional reduction

While the 6D theories provide a rich structure to study various techniques to construct scattering amplitudes, these amplitudes can also be used to study results in 4D theories by dimensional reduction. The map from the 6D $\mathcal{N}=(1,1)$ SYM to 4D $\mathcal{N}=4$ SYM is studied in [15]. Following their description, in this section, we will discuss the maps to get amplitudes in the half-maximal 4D theories from the 6D amplitudes we constructed in the previous subsections.

Before we derive the amplitudes, we will first introduce the 4D theories that arise from the dimensional reduction of the 6D $\mathcal{N}=(1,0)$ SQCD and $\mathcal{N}=(1,1)/(2,0)$ supergravity theories.

8.1 $\mathcal{N}$=2 SQCD

The half-maximal gauge theory with the eight supercharges in 4D is the $\mathcal{N}=2$ SQCD theory. This theory has the vector multiplet and its CPT conjugate,

$$\mathcal{V}_{+}^{\mathcal{N}=2} = A_{+} + \theta^{a}\lambda_{a} + (\theta)^{2}\phi \tag{8.1a}$$

$$\mathcal{V}_{-}^{\mathcal{N}=2} = \bar{\phi} + \theta^{a}\bar{\lambda}_{a} + (\theta)^{2}A_{-} \tag{8.1b}$$

where θ_a's are the fermionic superspace variable with $a = \{1, 2\}$, that have an internal R-symmetry $SU(2)$. The vector multiplets together have four bosonic DoF (2 vector bosons + 2 scalars) and four fermionic DoF (2 Weyl spinors). It also has a hypermultiplet given as

$$\Phi^{\mathcal{N}=2} = \phi + \theta^{a}\psi_{a} + (\theta)^{2}\tilde{\phi} \tag{8.2}$$

with two bosonic DoF (2 scalars) and two fermionic DoF (1 Weyl spinor).

Comparing with the multiplets from 6D $\mathcal{N} = (1, 0)$ SQCD, it is straightforward to see the mapping between the fields as

$$\mathcal{V}_{\dot{a}}^{\mathcal{N}=(1,0)} = \mathcal{V}_{+}^{\mathcal{N}=2} \oplus \mathcal{V}_{-}^{\mathcal{N}=2}, \tag{8.3a}$$

$$g_{a\dot{a}} \implies \{A_{+}, A_{-}, \phi, \bar{f}\}; \quad \{\tilde{\chi}_{\dot{a}}, \bar{\tilde{\chi}}_{\dot{a}}\} \implies \{\lambda_{a}, \bar{\lambda}_{a}\},$$

$$\Phi^{\mathcal{N}=(1,0)} = \Phi^{\mathcal{N}=2}. \tag{8.3b}$$

i.e., the $SU(2_L)$ little group symmetry in the 6D theory maps one to one as the $SU(2)$ R-symmetry in the 4D theory.

8.2 $\mathcal{N}$=4 Supergravity

The $\mathcal{N}=4$ supergravity theory has 16 supercharges. The theory contains a gravity multiplet and a vector multiplet as the matter multiplet. These can be expressed as

$$\mathcal{H}_{+}^{\mathcal{N}=4} = h_{+} + \theta^{A}\lambda_{A} + \theta^{A}\theta^{B}A_{[AB]}^{+} + \epsilon_{ABCD}\theta^{A}\theta^{B}\theta^{C}\lambda^{D} + (\eta)^{4}\phi, \tag{8.4a}$$

$$\mathcal{H}_{-}^{\mathcal{N}=4} = \bar{\phi} + \theta^{A}\bar{\lambda}_{A} + \theta^{A}\theta^{B}A_{[AB]}^{-} + \epsilon_{ABCD}\theta^{A}\theta^{B}\theta^{C}\bar{\lambda}^{D} + (\eta)^{4}h_{-}, \tag{8.4b}$$

$$\mathcal{V}^{\mathcal{N}=4} = A_{+} + \theta^{A}\lambda_{A} + \theta^{A}\theta^{B}\phi_{[AB]} + \epsilon_{ABCD}\theta^{A}\theta^{B}\theta^{C}\lambda^{D} + (\eta)^{4}A_{-} \tag{8.4c}$$

where θ_A's are the fermionic superspace variable with $A = \{1, 2, 3, 4\}$, that have an internal R-symmetry $SU(4)$. The graviton multiplets together have 16 bosonic DoF (2 graviton + 12 graviphoton+ 2 scalar) and 16 fermionic DoF (4 Dirac spinors), while the vector multiplet has eight bosonic (2 vector boson + 6 scalars) and eight fermionic DoF (4 Weyl spinors).

The 6D $\mathcal{N} = (2, 0)$ theory compactifies as

$$\mathcal{H}_{(\dot{a}\dot{b})}^{\mathcal{N}=(2,0)} = \mathcal{H}_{+}^{\mathcal{N}=4} \oplus \mathcal{H}_{-}^{\mathcal{N}=4} \oplus \mathcal{V}^{\mathcal{N}=4}, \tag{8.5a}$$

$$\mathcal{T}^{\mathcal{N}=(2,0)} = \mathcal{V}^{\mathcal{N}=4}. \tag{8.5b}$$

Starting with N_t matter multiples in 6D, we get $N_t + 1$ matter multiplets in 4D.

Similarly, for the $\mathcal{N} = (1, 1)$ theory,

$$\mathcal{H}_{a\dot{a}}^{\mathcal{N}=(1,1)} = \mathcal{H}_{+}^{\mathcal{N}=4} \oplus \mathcal{H}_{-}^{\mathcal{N}=4} \oplus \mathcal{V}^{\mathcal{N}=4} \oplus \mathcal{V}^{\mathcal{N}=4}, \tag{8.6a}$$

$$\mathcal{V}^{\mathcal{N}=(1,1)} = \mathcal{V}^{\mathcal{N}=4} \tag{8.6b}$$

with a total of $N_t + 2$ matter multiplets.

8.3 Massless 4D mapping

Now that we know how the fields map to each other, we can study the compactification of 6D amplitudes to 4D. Using the 6D amplitudes, we can derive maps for both the massless and massive 4D amplitudes. In this subsection, we will introduce the mapping for the massless case, before discussing the massive case in the next subsection.

To reduce to the massless subspace, for the choice of basis in 6D, we can make the special choice of setting $p_4 = p_5 = 0$. Then,

$$\begin{aligned} &p_{6\text{D}} = \{p_0, p_1, p_2, p_3, 0, 0\} \implies p_{4\text{D}} = \{p_0, p_1, p_2, p_3\} \\ &p_{6\text{D}}^2 = 0 \implies p_{4\text{D}}^2 \end{aligned} \tag{8.7}$$

Using the explicit parametrization shown in section 3, the 6D spinors are mapped to the 4D ones as

$$\lambda^{Aa} = \begin{pmatrix} 0 & \lambda_\alpha \\ \tilde{\lambda}^{\dot{\alpha}} & 0 \end{pmatrix}; \qquad \tilde{\lambda}_{A\dot{a}} = \begin{pmatrix} 0 & \lambda^\alpha \\ \tilde{\lambda}_{\dot{\alpha}} & 0 \end{pmatrix}, \tag{8.8}$$

and the momentum matrices are represented as

$$p_{AB} = \begin{pmatrix} 0 & -p^\alpha{}_{\dot{\beta}} \\ p_{\dot{\alpha}}{}^\beta & 0 \end{pmatrix}; \qquad p^{AB} = \begin{pmatrix} 0 & -p_\alpha{}^{\dot{\beta}} \\ p^{\dot{\alpha}}{}_\beta & 0 \end{pmatrix}. \tag{8.9}$$

Finally, for the SQCD theories in 6D and 4D, the supermomenta are mapped as,

$$q^A = \lambda^{Aa}\eta_a = \begin{pmatrix} \lambda_\alpha \theta^1 \\ \tilde{\lambda}^{\dot{\alpha}} \theta^2 \end{pmatrix} =: \begin{pmatrix} q^1_\alpha \\ q^{2\dot{\alpha}} \end{pmatrix} \tag{8.10}$$

where we have introduced the 4D supermomenta q^a_α with the Grassmann variables mapping as

$$\eta_a = (\theta^1, \theta^2). \tag{8.11}$$

4-point SQCD amplitudes

For illustrative purposes, we will derive the 4-point amplitudes in 4D $\mathcal{N} = 2$ SQCD using the results derived from section 5.6. All the amplitudes($\geq$ 4-point) contain supermomentum conservation as a prefactor. This delta function maps as

$$\begin{aligned} \delta^{(4)}(Q_{6\text{D}}) =& \frac{1}{4!}\epsilon_{ABCD} Q^A_{(6\text{D})} Q^B_{(6\text{D})} Q^C_{(6\text{D})} Q^D_{(6\text{D})} = Q^1_{(6\text{D})} Q^2_{(6\text{D})} Q^3_{(6\text{D})} Q^4_{(6\text{D})} \\ =& Q_1{}^1_{(4\text{D})} Q_1{}^2_{(4\text{D})} Q_2{}^1_{(4\text{D})} Q_2{}^2_{(4\text{D})}; \quad [Q^\alpha_{m(4\text{D})} := \sum_i q^\alpha_{mi}] \\ =& \frac{1}{4}(\epsilon_{\alpha\beta} Q_1{}^\alpha_{(4\text{D})} Q_1{}^\beta_{(4\text{D})})(\epsilon_{\dot{\alpha}\dot{\beta}} Q_2{}^{\dot{\alpha}}_{(4\text{D})} Q_2{}^{\dot{\beta}}_{(4\text{D})}) \\ =& \delta^{(2)}(Q_{1(4\text{D})})\delta^{(2)}(Q_{2(4\text{D})}) := \delta^{(4)}(Q_{(4\text{D})}). \end{aligned} \tag{8.12}$$

Now we can reduce the amplitudes to 4D. Starting with the two vector two hyper amplitude (4.22),

$$\mathcal{A}_{6\text{D}}(\mathcal{V}_{1\dot{a}}, \mathcal{V}_{2\dot{b}}, \Phi_3, \Phi_4) \implies \mathcal{A}_{4\text{D}}(\mathcal{V}_{1\pm}, \mathcal{V}_{2\pm}, \Phi_3, \Phi_4)$$

$$
\begin{aligned}
&s_{12} \implies s_{12}; \quad s_{14} \implies s_{14} \\
&\delta^{(4)}(Q_{(6\mathrm{D})}) \implies \delta^{(4)}(Q_{(4\mathrm{D})}) \\
&[1_{\dot{a}}|\not{p}_3|2_{\dot{b}}] \implies \begin{pmatrix} 0 & \lambda_1^{\alpha} \\ \tilde{\lambda}_{1\dot{\alpha}} & 0 \end{pmatrix}^T \begin{pmatrix} 0 & -p_{3\alpha}{}^{\dot{\beta}} \\ p_3{}^{\dot{\alpha}}{}_{\beta} & 0 \end{pmatrix} \begin{pmatrix} 0 & \lambda_2^{\beta} \\ \tilde{\lambda}_{2\dot{\beta}} & 0 \end{pmatrix} = \begin{pmatrix} 0 & [13]\langle 32\rangle \\ \langle 13\rangle[23] & 0 \end{pmatrix}
\end{aligned}
$$

$$
\implies \mathcal{A}_{4\mathrm{D}}(\mathcal{V}_{1-}, \mathcal{V}_{2+}, \Phi_3, \Phi_4) = \Omega_{f_3 f_4}\delta^{(4)}(Q)\frac{[13]\langle 32\rangle}{s_{12}s_{14}} \tag{8.13a}
$$

$$
\mathcal{A}_{4\mathrm{D}}(\mathcal{V}_{1+}, \mathcal{V}_{2-}, \Phi_3, \Phi_4) = \Omega_{f_3 f_4}\delta^{(4)}(Q)\frac{\langle 13\rangle[23]}{s_{12}s_{14}} \tag{8.13b}
$$

Following this prescription, any 6D amplitude can be reduced to a massless 4D amplitude.

8.4 Massive 4D mapping

In the massive cases, instead of setting the special subspace to $\{0, 0\}$, we will now use them to represent the masses of the particles:

$$
\begin{aligned}
&p_{6\mathrm{D}} = \{p_0, p_1, p_2, p_3, p_4, p_5\} \implies p_{4\mathrm{D}} = \{p_0, p_1, p_2, p_3\}, \\
&p_{6\mathrm{D}}^2 = 0 \implies p_{4\mathrm{D}}^2 = |m|^2, && (8.14a) \\
&|m|^2 = m\bar{m}, \ p_5 + ip_4 = m. && (8.14b)
\end{aligned}
$$

The map for the momentum matrices is

$$
p_{AB} = \begin{pmatrix} -\bar{m}\epsilon^{\alpha\beta} & -p^{\alpha}{}_{\dot{\beta}} \\ p_{\dot{\alpha}}{}^{\beta} & m\epsilon_{\dot{\alpha}\dot{\beta}} \end{pmatrix}, \qquad p^{AB} = \begin{pmatrix} m\epsilon_{\alpha\beta} & -p_{\alpha}{}^{\dot{\beta}} \\ p^{\dot{\alpha}}{}_{\beta} & -\bar{m}\epsilon^{\dot{\alpha}\dot{\beta}} \end{pmatrix}. \tag{8.15}
$$

A massive 4D momentum is represented as a bispinor that satisfies

$$
\begin{aligned}
&p^{\mu}\sigma_{\mu\alpha\dot{\alpha}} = p_{\alpha\dot{\alpha}} = \lambda_{\alpha}\tilde{\lambda}_{\dot{\alpha}} + \mu_{\alpha}\tilde{\mu}_{\dot{\alpha}} \\
&\langle\lambda\mu\rangle = m; \quad [\tilde{\lambda}\tilde{\mu}] = \bar{m}
\end{aligned} \tag{8.16}
$$

For our particular parametrization, the spinors reduce as

$$
\lambda^{Aa} = \begin{pmatrix} -\mu_{\alpha} & -\lambda_{\alpha} \\ \tilde{\lambda}_{\dot{\alpha}} & \tilde{\mu}_{\dot{\alpha}} \end{pmatrix}, \quad \tilde{\lambda}_{A\dot{a}} = \begin{pmatrix} \frac{\bar{m}}{m}\mu_{\alpha} & -\lambda_{\alpha} \\ \lambda_{\dot{\alpha}} & -\frac{m}{\bar{m}}\tilde{\mu}_{\dot{\alpha}} \end{pmatrix}. \tag{8.17}
$$

For the SQCD theories, the supermomenta map the same as the massless case Eq.(8.10), with the redefinition of 4D supermomenta as

$$
\begin{aligned}
&q^A = \begin{pmatrix} q_{\alpha}^1 \\ q^{2\dot{\alpha}} \end{pmatrix}, \\
&q_{\alpha}^1 = \lambda_{\alpha}\theta^1 - \mu_{\alpha}\theta^2, \quad q_{\dot{\alpha}}^2 = \tilde{\lambda}_{\dot{\alpha}}\theta^2 - \tilde{\mu}_{\dot{\alpha}}\theta^1. && (8.18a)
\end{aligned}
$$

For the supergravity theories, the Grassmann variables map as

$$
\begin{aligned}
&\mathcal{N} = (2,0)_{6\mathrm{D}} \implies \mathcal{N} = 4_{4\mathrm{D}} : (\eta_{\alpha}^a) \implies (\theta^A), && (8.19a) \\
&\mathcal{N} = (1,1)_{6\mathrm{D}} \implies \mathcal{N} = 4_{4\mathrm{D}} : (\eta^a, \tilde{\eta}^{\dot{a}}) \implies (\theta^A). && (8.19b)
\end{aligned}
$$

The massive supercharges in 4D supergravity are given as

$$
\begin{aligned}
&q_{\alpha}^1 = \lambda_{\alpha}\theta^1 - \mu_{\alpha}\theta^2, \quad q_{\dot{\alpha}}^2 = \tilde{\lambda}_{\dot{\alpha}}\theta^2 - \tilde{\mu}_{\dot{\alpha}}\theta^1, \\
&q_{\alpha}^3 = \lambda_{\alpha}\theta^3 - \mu_{\alpha}\theta^4, \quad q_{\dot{\alpha}}^4 = \tilde{\lambda}_{\dot{\alpha}}\theta^4 - \tilde{\mu}_{\dot{\alpha}}\theta^3.
\end{aligned} \tag{8.20}
$$

5-Point supergravity amplitudes

Here we derive the massive 5-point amplitudes in 4D supergravity from the 6D supergravity theories. For the flavor choice $\{1,2\},\{3,4\}$, let the mass of each flavor be denoted as m_1, m_3 respectively. Then the Mandelstam variables map as,

$$\begin{aligned} &s_{15} = s_{15} - m_1^2; \quad s_{25} = s_{25} - m_1^2; \quad s_{45} = s_{45} - m_3^2; \quad s_{35} = s_{35} - m_3^2 \\ &s_{12} = s_{12}; \quad s_{34} = s_{34} \end{aligned} \tag{8.21}$$

Similar to the massless case, the supermomentum conservation maps as,

$$\begin{aligned} &\mathcal{N} = (2,0)_{6\text{D}} \implies \mathcal{N} = 4_{4\text{D}} : \\ &\prod_{\alpha=1,2} \delta^{(4)}(Q_{\alpha 6D}) = Q_{1\,(4\text{D})}^{1} Q_{1\,(4\text{D})}^{2} Q_{2\,(4\text{D})}^{1} Q_{2\,(4\text{D})}^{2} Q_{3\,(4\text{D})}^{1} Q_{3\,(4\text{D})}^{2} Q_{4\,(4\text{D})}^{1} Q_{4\,(4\text{D})}^{2} \\ \implies &\frac{1}{(2!)^4} \prod_{A=1}^{4} \delta^{(2)}(Q_{(4\text{D})}^{A}) := \delta^{(8)}(Q_{4\text{D}}) \end{aligned} \tag{8.22}$$

$$\begin{aligned} &\mathcal{N} = (\mathbf{1},\mathbf{1})_{6\text{D}} \implies \mathcal{N} = \mathbf{4}_{4\text{D}} : \\ \implies &\delta^{(4)}(Q_{6\text{D}})\delta^{(4)}(\tilde{Q}_{6\text{D}}) := \delta^{(8)}(Q_{4\text{D}}) \end{aligned} \tag{8.23}$$

Finally, the numerators in the f_α functions in the half-maximal chiral supergravity map as

$$\begin{aligned} &\mathcal{N} = (2,0)_{6\text{D}} \implies \mathcal{N} = 4_{4\text{D}} : \\ &\langle q_{\alpha i}|5_{\dot{a}}] = \begin{pmatrix} \langle q_i^1 5\rangle & \langle q_i^3 5\rangle \\ [q_i^2 5] & [q_i^4 5] \end{pmatrix}, \\ &\langle q_{\alpha 5}|\not{p}_i\not{p}_j|5_{\dot{a}}] = \begin{pmatrix} -\langle q_5^1 p_i p_j 5\rangle - m_j[q_5^2 p_i 5\rangle + m_i[q_5^2 p_j 5\rangle & -\langle q_5^3 p_i p_j 5\rangle - m_j[q_5^4 p_i 5\rangle + m_i[q_5^4 p_j 5\rangle \\ -\bar{m}_i\langle q_5^1 p_j 5] + \bar{m}_j\langle q_5^1 p_i 5] + [q_5^2 p_i p_j 5] & -\bar{m}_i\langle q_5^3 p_j 5] + \bar{m}_j\langle q_5^3 p_i 5] + [q_5^4 p_i p_j 5] \end{pmatrix}. \end{aligned} \tag{8.24}$$

Similarly for the non-chiral supergravity

$$\begin{aligned} &\mathcal{N} = (1,1)_{6\text{D}} \implies \mathcal{N} = 4_{4\text{D}} : \\ &\langle q_i|5_{\dot{a}}] = \begin{pmatrix} \langle q_i^1 5\rangle \\ [q_i^2 5] \end{pmatrix}; \quad [\tilde{q}_i|5_a\rangle = \begin{pmatrix} \langle q_i^3 5\rangle \\ [q_i^4 5] \end{pmatrix}, \\ &\langle q_5|\not{p}_i\not{p}_j|5_{\dot{a}}] = \begin{pmatrix} -\langle q_5^1 p_i p_j 5\rangle - m_j[q_5^2 p_i 5\rangle + m_i[q_5^2 p_j 5\rangle \\ -\bar{m}_i\langle q_5^1 p_j 5] + \bar{m}_j\langle q_5^1 p_i 5] + [q_5^2 p_i p_j 5] \end{pmatrix}, \\ &[\tilde{q}_5|\not{p}_i\not{p}_j|5_a\rangle = \begin{pmatrix} -[5 p_j p_i q_5^4] - m_i[5 p_j q_5^3\rangle + m_j[5 p_i q_5^3\rangle \\ -\bar{m}_j\langle 5 p_i q_5^4] + \bar{m}_i\langle 5 p_j q_5^4] + \langle 5 p_j p_i q_5^3\rangle \end{pmatrix}. \end{aligned} \tag{8.25}$$

Using the mapping for superfields Eq.(8.5) and Eq.(8.6), the superamplitudes map is given by

$$\mathcal{M}^{\mathcal{N}=(2,0)}(\mathcal{T}_1,\mathcal{T}_2,\mathcal{T}_3,\mathcal{T}_4,\mathcal{H}_{5_{11}}) \implies \mathcal{M}^{\mathcal{N}=4}(\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3,\mathcal{V}_4,\mathcal{H}_{5_+}), \tag{8.26a}$$

$$\mathcal{M}^{\mathcal{N}=(2,0)}(\mathcal{T}_1,\mathcal{T}_2,\mathcal{T}_3,\mathcal{T}_4,\mathcal{H}_{5_{22}}) \implies \mathcal{M}^{\mathcal{N}=4}(\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3,\mathcal{V}_4,\mathcal{H}_{5_-}), \tag{8.26b}$$

$$\mathcal{M}^{\mathcal{N}=(1,1)}(\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3,\mathcal{V}_4,\mathcal{H}_{5_{12}}) \implies \mathcal{M}^{\mathcal{N}=4}(\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3,\mathcal{V}_4,\mathcal{H}_{5_+}), \tag{8.26c}$$

$$\mathcal{M}^{\mathcal{N}=(1,1)}(\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3,\mathcal{V}_4,\mathcal{H}_{5_{21}}) \implies \mathcal{M}^{\mathcal{N}=4}(\mathcal{V}_1,\mathcal{V}_2,\mathcal{V}_3,\mathcal{V}_4,\mathcal{H}_{5_-}), \tag{8.26d}$$

The final expression of the amplitudes can be derived by substituting Eq.(8.25) and Eq.(8.24) into Eq.(7.15) and Eq.(7.18), respectively.

9 Conclusion

This thesis was focused on constructing superamplitudes in the 6D $\mathcal{N} = (1, 0)$ SQCD theory and 6D $\mathcal{N} = (1, 1)/\mathcal{N} = (2, 0)$ supergravity theories using 6D spinor helicity, the bootstrap approach, and the double copy structure and mapping them to 4D $\mathcal{N} = 2$ SQCD theory and 4D $\mathcal{N} = 4$ supergravity, respectively using dimensional reduction.

We initially reviewed 6D spinor helicity and supersymmetry in 6D to construct the 3,4-point superamplitudes in the 6D SQCD theory. These results can be verified by comparing them with the construction of 6D $\mathcal{N} = (1, 1)$ superamplitudes [14] and 5D superamplitudes [7]. An alternate form for the 3-point superamplitudes that separates the chiral and anti-chiral components is also derived. The main result in the thesis was to construct the 5-point amplitude with one vector and four hyper massless states in the 6D SQCD theory using the bootstrap approach. This amplitude was verified using factorization to lower-point amplitudes.

We then used the double copy structure to construct the 3,4,5-point amplitudes in the 6D supergravity theories. Finally, following [15], we introduced the maps from 6D $\mathcal{N} = (1, 0)$ SQCD to 4D $\mathcal{N} = 2$ SQCD and 6D $\mathcal{N} = (1, 1)/\mathcal{N} = (2, 0)$ supergravity to 4D $\mathcal{N} = 4$ supergravity. These maps can be used to construct the massless and massive amplitudes in the 4D theories. We also verified that the map by checking massless 4-point amplitudes against known results in $\mathcal{N} = 2$ SQCD [38].

Building upon this thesis, as future work, one can construct amplitudes to higher multiplicities using the bootstrap approach introduced in the thesis. The bottleneck to this approach is presumed to be finding the basis function f's that satisfy the supersymmetric Ward identities. While the 5-point f's were simple, the ansatz for finding them at higher multiplicities could be more involved due to the increasing number of linearly independent numerators.

Another promising direction to extend this thesis is to study the classical limit of the massive supergravity amplitudes computed in section 8. The four external scalars can be used to model massive objects (such as black holes or neutron stars) flying past each other under gravity. Then, following [15], one can get the leading-order waveform of the radiation as a function of the graviton momentum. With the upcoming improvements to LIGO, increased precision of such waveforms would be required to match the sensitive detectors.

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Appendix A: Conventions

In this appendix, we define the conventions used in this thesis. We follow the conventions used in [15]. We work with the mostly negative signature for the metric. The sigma matrices satisfy the Clifford algebra,

$$\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = 2\eta^{\mu\nu} \tag{A.1}$$

The 4D Pauli matrices are given as,

$$\begin{aligned} \sigma_0 &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \sigma_2 &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned} \tag{A.2}$$

Then the 4D sigma matrices are

$$\sigma^\mu = (\sigma_0, \sigma_1, \sigma_2, \sigma_3); \quad \bar{\sigma}^\mu = (\sigma_0, -\sigma_1, -\sigma_2, -\sigma_3) \tag{A.3}$$

The $SU(2)$ Levi-Cevita is defines as

$$\epsilon_{\alpha\beta} = \epsilon_{\dot{\alpha}\dot{\beta}} = -\epsilon^{\alpha\beta} = -\epsilon^{\dot{\alpha}\dot{\beta}} = i\sigma_2 \tag{A.4}$$

The 6D sigma matrices are defined as,

$$\Sigma^0 = i\sigma_1 \otimes \sigma_2; \qquad \bar{\Sigma}^0 = -\Sigma^0; \tag{A.5a}$$
$$\Sigma^1 = i\sigma_2 \otimes \sigma_3; \qquad \bar{\Sigma}^0 = \Sigma^1; \tag{A.5b}$$
$$\Sigma^2 = -\sigma_2 \otimes \sigma_0; \qquad \bar{\Sigma}^0 = -\Sigma^2; \tag{A.5c}$$
$$\Sigma^3 = -i\sigma_2 \otimes \sigma_1; \qquad \bar{\Sigma}^0 = \Sigma^3; \tag{A.5d}$$
$$\Sigma^4 = -\sigma_3 \otimes \sigma_2; \qquad \bar{\Sigma}^0 = -\Sigma^4; \tag{A.5e}$$
$$\Sigma^5 = i\sigma_0 \otimes \sigma_2; \qquad \bar{\Sigma}^0 = \Sigma^5; \tag{A.5f}$$

The $SU(4)$ Levi-Cevita are defined with

$$\epsilon^{1234} = \epsilon_{1234} = 1 \tag{A.6}$$

The 6D sigma matrices satisfy the properties,

$$\Sigma^\mu_{AB} = \frac{1}{2}\epsilon_{ABCD}\bar{\Sigma}^{CD} \tag{A.7a}$$
$$\Sigma^\mu_{AB}\Sigma_{\mu CD} = -2\epsilon_{ABCD} \tag{A.7b}$$
$$\Sigma^\mu_{AB}\bar{\Sigma}^{CD}_\mu = -2(\delta^A_C\delta^B_D - \delta^B_C\delta^A_D) \tag{A.7c}$$
$$\mathrm{Tr}(\bar{\Sigma}^\mu\Sigma^\nu) = 4\eta^{\mu\nu} \tag{A.7d}$$

Appendix B: 4-point BCFW tensor contraction

This appendix follows the derivation given in [8, 14].To simplify the tensor contraction $\Gamma_{ade}\Gamma_{b\ c}^{\ e}$ further, we need to compute $u_{pe}u^{e}_{-p}\tilde{u}_{p\dot{e}}\tilde{u}^{\dot{e}}_{-p}$. Consider the following object,

$$\begin{aligned}
\langle\hat{1}_a|p_4p_{\hat{2}}|\hat{1}_{\dot{a}}] =&\langle\hat{1}_a|4_{\dot{d}}][4^{\dot{d}}|p_{\hat{2}}|\hat{1}_{\dot{a}}]\\
u_{1a}\tilde{u}_{4\dot{d}}[4^{\dot{d}}|p_{\hat{2}}|\hat{1}_{\dot{a}}] =&u_{1a}\tilde{u}_{4\dot{d}}[4^{\dot{d}}|p_{\hat{2}}|\hat{1}_{\dot{a}}]\\
u_{1a}\tilde{u}_{p\dot{e}}[p^{\dot{e}}|p_{\hat{2}}|\hat{1}_{\dot{a}}] =&u_{1a}\tilde{u}_{\hat{1}\dot{d}}[\hat{1}^{\dot{d}}|p_{\hat{2}}|\hat{1}_{\dot{a}}]\\
iu_{1a}\tilde{u}_{p\dot{e}}[(-p)^{\dot{e}}|p_{\hat{2}}|\hat{1}_{\dot{a}}] =&u_{1a}\tilde{u}_{1\dot{d}}[1^{\dot{d}}|\hat{2}^b\rangle\langle\hat{2}_b|\hat{1}_{\dot{a}}]\\
iu_{1a}\tilde{u}_{p\dot{e}}[(-p)^{\dot{e}}|\hat{2}^b\rangle\langle\hat{2}_b|\hat{1}_{\dot{a}}] =&u_{1a}\tilde{u}_{1\dot{d}}\det(\langle\hat{2}_b|\hat{1}_{\dot{a}}])\delta^{\dot{d}}_{\dot{a}}\\
-u_{1a}\tilde{u}_{p\dot{e}}\tilde{u}^{\dot{e}}_{-p}u^{e}_{-p}\langle p_e|\hat{1}_{\dot{a}}] =&u_{1a}\tilde{u}_{1\dot{d}}s_{\hat{1}\hat{2}}\delta^{\dot{d}}_{\dot{a}}\\
-u_{1a}\tilde{u}_{1\dot{a}}(u_{pe}u^{e}_{-p}\tilde{u}_{p\dot{e}}\tilde{u}^{\dot{e}}_{-p}) =&u_{1a}\tilde{u}_{1\dot{a}}s_{12}\\
\implies u_{pe}u^{e}_{-p}\tilde{u}_{p\dot{e}}\tilde{u}^{\dot{e}}_{-p} =&-s_{12}
\end{aligned}\tag{B.1}$$

Also using b shifts, we can always choose w and $\tilde{w}$ such that,

$$u_{pe}w^{e}_{k} = \tilde{u}_{p\dot{e}}\tilde{w}^{\dot{e}}_{k} = 0\tag{B.2}$$

Finally, from the definition, $u_{pa}w_{pb} - u_{pb}w_{pa} = \epsilon_{ab}$,

$$\begin{aligned}
&(u_{pa}w_{pb} - u_{pb}w_{pa})(u^a_pw^b_p - u^b_pw^a_p) = \epsilon_{ab}\epsilon^{ab}\\
\implies&(u_{pa}w_{pb} - u_{pb}w_{pa})(iu^a_{-p}iw^b_{-p} - iu^b_{-p}iw^a_{-p}) = -2\\
\implies&u_{pa}u^a_{-p} = \frac{1}{w_{pa}w^a_{-p}}
\end{aligned}\tag{B.3}$$

Then, we can simplify the tensorial contractions as

$$\begin{aligned}
\Gamma_{ade}\Gamma_{b\ c}^{\ e} =&(u_{\hat{1}a}u_{4d}w_{pe} + u_{\hat{1}a}w_{4d}u_{pe} + w_{\hat{1}a}u_{4d}u_{pe})(u_{\hat{2}b}u^{e}_{-p}w_{3c} + u_{\hat{2}b}w^{e}_{-p}u_{3c} + w_{\hat{2}b}u^{e}_{-p}u_{3c})\\
=&\frac{1}{\tilde{u}_{p\dot{e}}\tilde{u}^{\dot{e}}_{-p}}\big(u_{\hat{1}a}u_{4d}u_{\hat{2}b}u_{3c} - su_{\hat{1}a}w_{4d}u_{\hat{2}b}w_{3c} - su_{\hat{1}a}w_{4d}w_{\hat{2}b}u_{3c}\\
&- sw_{\hat{1}a}u_{4d}u_{\hat{2}b}w_{3c} - sw_{\hat{1}a}u_{4d}w_{\hat{2}b}u_{3c}\big)
\end{aligned}\tag{B.4}$$

Projecting $\langle\hat{1}_a\hat{2}_b3_c4_d\rangle$ in u_i, w_i basis, we note that

$$\begin{aligned}
&u^a_1u^b_2\langle\hat{1}_a\hat{2}_b3_c4_d\rangle && w^a_1w^b_2\langle\hat{1}_a\hat{2}_b3_c4_d\rangle\\
=&iu^a_pu^b_{-p}\langle p_ap_b3_c4_d\rangle &=&\langle(w_p.p + w_4.4)(iw_{-p}.p + w_3.3)3_c4_d\rangle\\
=&-iu_p.u_{-p}\langle3_c|\not{p}|4_d\rangle &=&-iw_p.w_{-p}\langle3_c|\not{p}|4_d\rangle - w_{3c}w_{4d}(2p_3\cdot p_4)\\
=&u_p.u_{-p}\tilde{u}_p.\tilde{u}_{-p}u_{3c}u_{4d} &=&w_p.w_{-p}\tilde{u}_p.\tilde{u}_{-p}u_{3c}u_{4d} - sw_{3c}w_{4d}\\
=&-su_{3c}u_{4d}, &=&u_{3c}u_{4d} - sw_{3c}w_{4d}.
\end{aligned}\tag{B.5}$$

Projecting into other u_i, w_i, we get

$$\begin{aligned}
\langle\hat{1}_a\hat{2}_b3_c4_d\rangle =&u_{\hat{1}a}u_{4d}u_{\hat{2}b}u_{3c} - su_{\hat{1}a}w_{4d}u_{\hat{2}b}w_{3c} - su_{\hat{1}a}w_{4d}w_{\hat{2}b}u_{3c}\\
&- sw_{\hat{1}a}u_{4d}u_{\hat{2}b}w_{3c} - sw_{\hat{1}a}u_{4d}w_{\hat{2}b}u_{3c}.
\end{aligned}\tag{B.6}$$

Similarly for the anti-chiral tensor,

$$\Gamma_{\dot{a}\dot{d}\dot{e}}\Gamma_{\dot{b}\ \dot{c}}^{\ \dot{e}} = \frac{1}{u_{pe}u^{e}_{-p}}[\hat{1}_{\dot{a}}\hat{2}_{\dot{b}}3_{\dot{c}}4_{\dot{d}}]\tag{B.7}$$

Bibliography

[1] H. Elvang and Y. tin Huang, *Scattering amplitudes*, 2014. [1308.1697].

[2] P. Benincasa and F. Cachazo, *Consistency conditions on the s-matrix of massless particles*, 2008. [0705.4305].

[3] M.L. Mangano and S.J. Parke, *Multi-parton amplitudes in gauge theories*, *Physics Reports* **200** (1991) 301–367 [hep-th/0509223].

[4] L. Dixon, *Calculating scattering amplitudes efficiently*, 1996. [hep-ph/9601359].

[5] Z. Xu, D.-H. Zhang and L. Chang, *Helicity Amplitudes for Multiple Bremsstrahlung in Massless Nonabelian Gauge Theories*, *Nucl. Phys. B* **291** (1987) 392.

[6] S. Jain, R.R. John, A. Mehta, A.A. Nizami and A. Suresh, *Higher spin 3-point functions in 3d cft using spinor-helicity variables*, *Journal of High Energy Physics* **2021** (2021) [2106.00016].

[7] M. Chiodaroli, M. Günaydin, H. Johansson and R. Roiban, *Spinor-helicity formalism for massive and massless amplitudes in five dimensions*, *Journal of High Energy Physics* **2023** (2023) [2202.08257].

[8] C. Cheung and D. O'Connell, *Amplitudes and spinor-helicity in six dimensions*, *Journal of High Energy Physics* **2009** (2009) 075–075 [0902.0981].

[9] S. Caron-Huot and D. O'Connell, *Spinor helicity and dual conformal symmetry in ten dimensions*, *Journal of High Energy Physics* **2011** (2011) [1010.5487].

[10] N. Arkani-Hamed, T.-C. Huang and Y. tin Huang, *Scattering amplitudes for all masses and spins*, 2021. [1709.04891].

[11] R. Britto, F. Cachazo, B. Feng and E. Witten, *Direct proof of the tree-level scattering amplitude recursion relation in yang-mills theory*, *Physical Review Letters* **94** (2005) [hep-th/0501052].

[12] R. Roiban, *Review of ads/cft integrability, chapter v.1: Scattering amplitudes: A brief introduction*, *Letters in Mathematical Physics* **99** (2011) 455–479 [1012.4001].

[13] M.F. Sohnius and P.C. West, *Conformal Invariance in N=4 Supersymmetric Yang-Mills Theory*, *Phys. Lett. B* **100** (1981) 245.

[14] T. Dennen, Y.-t. Huang and W. Siegel, *Supertwistor space for 6d maximal super yang-mills*, *Journal of High Energy Physics* **2010** (2010) .

[15] J. Plefka, T. Schuster and V. Verschinin, *From six to four and more: massless and massive maximal super yang-mills amplitudes in 6d and 4d and their hidden symmetries*, *Journal of High Energy Physics* **2015** (2015) [1405.7248].

[16] N. Bjerrum-Bohr and P. Vanhove, *Explicit cancellation of triangles in one-loop gravity amplitudes*, *Journal of High Energy Physics* **2008** (2008) 065–065 [0802.0868].

[17] Z. Bern, J.J. Carrasco, M. Chiodaroli, H. Johansson and R. Roiban, *The duality between color and kinematics and its applications*, 2019. [1909.01358].

[18] H. Kawai, D.C. Lewellen and S.H.H. Tye, *A Relation Between Tree Amplitudes of Closed and Open Strings*, *Nucl. Phys. B* **269** (1986) 1.

[19] Y.-t. Huang, H. Johansson, M. Santagata and C. Wen, *Surprising one-loop finiteness of 6d half-maximal supergravities*, *Phys. Rev. Lett.* **136** (2026) 111601 [2510.24558].

[20] H. Johansson, G. Kälin and G. Mogull, *Two-loop supersymmetric qcd and half-maximal supergravity amplitudes*, *Journal of High Energy Physics* **2017** (2017) [1706.09381].

[21] N. Seiberg and E. Witten, *Monopoles, duality and chiral symmetry breaking in n = 2 supersymmetric qcd*, *Nuclear Physics B* **431** (1994) 484–550 [hep-th/9408099].

[22] S. Weinberg, *The Quantum Theory of Fields, Volume 1: Foundations*, Cambridge University Press, Cambridge, UK (1995).

[23] R.J. Eden, P.V. Landshoff, D.I. Olive and J.C. Polkinghorne, *The Analytic S-Matrix*, Cambridge University Press, Cambridge, UK (1966).

[24] L. Bhardwaj, *Classification of 6d N=(1,0) gauge theories*, *JHEP* **11** (2015) 002 [1502.06594].

[25] N. Seiberg, *Notes on theories with 16 supercharges*, *Nucl. Phys. B Proc. Suppl.* **67** (1998) 158 [hep-th/9705117].

[26] H. Nishino and E. Sezgin, *The Complete N=2, D=6 Supergravity With Matter and Yang-Mills Couplings*, *Nucl. Phys. B* **278** (1986) 353.

[27] E. Malek, *Half-maximal supersymmetry from exceptional field theory*, *Fortsch. Phys.* **65** (2017) 1700061 [1707.00714].

[28] R. D'Auria, S. Ferrara and C. Kounnas, *N=(4,2) chiral supergravity in six-dimensions and solvable Lie algebras*, *Phys. Lett. B* **420** (1998) 289 [hep-th/9711048].

[29] Y. Tanii, *N=8 Supergravity in Six-Dimensions*, *Phys. Lett. B* **145** (1984) 197.

[30] Y. Bertrand, S. Hohenegger, O. Hohm and H. Samtleben, *Toward Exotic 6D Supergravities*, *JHEP* **01** (2021) 017 [2007.11644].

[31] B. de Wit, *Supergravity*, 2002. [hep-th/0212245].

[32] V.P. Nair, *A current algebra for some gauge theory amplitudes*, *Physics Letters B* **214** (1988) 215.

[33] H. Elvang, D.Z. Freedman and M. Kiermaier, *Susy ward identities, superamplitudes and counterterms*, *Journal of Physics A: Mathematical and Theoretical* **44** (2011) 454009 [1012.3401].

[34] R. Kleiss and H. Kuijf, *Multigluon cross sections and 5-jet production at hadron colliders*, *Nuclear Physics B* **312** (1989) 616.

[35] Z. Bern, J.J.M. Carrasco and H. Johansson, *New relations for gauge-theory amplitudes*, *Phys. Rev. D* **78** (2008) 085011 [0805.3993].

[36] Z. Bern, J.J.M. Carrasco and H. Johansson, *Perturbative quantum gravity as a double copy of gauge theory*, *Phys. Rev. Lett.* **105** (2010) 061602 [1004.0476].

[37] H. Johansson and A. Ochirov, *Double copy for massive quantum particles with spin*, *Journal of High Energy Physics* **2019** (2019) [1906.12292].

[38] H. Johansson and A. Ochirov, *Pure gravities via color-kinematics duality for fundamental matter*, *Journal of High Energy Physics* **2015** (2015) [1407.4772].